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Spatial Learning in Rats

By

Benbow Ferguson Ritchie
A.B. (University of Wisconsin) 1937
M.A. (University of California) 1945

DISSERTATION

Submitted in partial satisfaction of the requirements for the degree of

DOCTOR OF PHILOSOPHY

in

Psychology

in the

GRADUATE DIVISION

of the

UNIVERSITY OF CALIFORNIA

Approved:

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UNIVERSITY OF CALIFORNIA
GRADUATE DIVISION, NORTHERN SECTION

SUMMARY OF THE DISSERTATION

SUBMITTED IN PARTIAL SATISFACTION
OF THE REQUIREMENTS FOR THE

DEGREE OF DOCTOR OF PHILOSOPHY

BY

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A.B. (University of Wisconsin) 1937
M.A. (University of California) 1945

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Spatial Learning in Rats

The present study begins with a logical analysis of the facts of simple conditioning and of simple maze learning. This analysis provides (1) a way of describing the learning process which is more precise than the orthodox description in terms of stimulus-response concepts, and (2) a set of assumptions which will explain the results of several different simple maze experiments.

Two theorems entailed by these assumptions are tested experimentally.

Theorem I. A group of rats trained from S_1 to F_1 and S_2 to F_1, will learn faster than a group trained from S_1 to F_1 and S_3 to F_3.

The experimental evidence presented indicates that Theorem I is probably true.

Theorem II. A group of rats trained from S_1 to F_1 will have a disposition to take the shortest path from S_2 to F_1.

The experimental evidence presented indicates that Theorem II is probably false.

In conclusion a method of revising these assumptions is suggested. This revision changes the assumptions so that no theorems known to be false are entailed.

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A framework for an empirical ethics (with A. Kaplan). *Phil. Sci.*, 7, 476–491, 1940.

Correlations between VTE's on a maze and on a visual discrimination apparatus (with E. C. Tolman). *Jour. Comp. Psychol.*, 36, 91–98, 1943.

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Studies in spatial learning. IV. The transfer of place learning to other starting paths (with E. C. Tolman and D. Kalish). *Jour. Exp. Psychol.* (in press).

Take care of the sense and sounds
will take care of themselves.

--L. Carroll

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PART I

INTRODUCTION - THE LOGICAL ANALYSIS

INTRODUCTION - THE LOGICAL ANALYSIS

1. The S-R Hypothesis

> The essential nature of the learning process may be ... stated quite simply. Just as the inherited equipment of reaction tendencies consists of receptor-effector connections, so the process of learning consists in the strengthening of certain connections as contrasted with others, or in setting up quite new connections. (8)

This quotation from Hull will serve to illustrate what I shall call the "S-R hypothesis". Learning according to this hypothesis is a process in which certain stimulus response connections are either created or strengthened. According to Thorndike these connections are to be called "stimulus response bonds" while other psychologists use a different vocabulary. But in general most American psychologists maintain the S-R hypothesis, agreeing that learning is a process in which such S-R connections are modified.

One might conclude from this almost universal agreement among psychologists that there must be a great deal of evidence in favor of the S-R hypothesis. But this, I think, would be an error. Let us consider for a moment what is meant by the expression "stimulus response connection". It is obvious that until we know what is meant by

this, our search for evidence will be futile. But the advocates of the S-R hypothesis have never given a clear answer to this apparently impertinent question. In general they answer it either with vague speculations about the neural connection supposedly involved in the connection, or with S-R diagrams in which the letter "S" (the name of a stimulus) is connected by an arrow with the letter "R" (the name of a response). And in the latter case we are never told what it is that is named by the arrow. On the other hand, to the degree that the neural speculations are sufficiently specific to be testable, most of the evidence, discovered for example by Lashley (9), Loucks (11), and others, is against the S-R hypothesis.[1]

Now it is not my contention that the S-R hypothesis cannot be made intelligible, but merely that no one has as yet taken the trouble to do so. Since I am concerned in this introduction with stating as clearly as I can the differences between the opinions of the S-R psychologists and those of the Tolman school,[2] it is important

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1. This may account for the fact that such neural speculations are today considered somewhat <u>outré</u> by such an S-R advocate as Spence (17).

2. By the "views of the Tolman school" I mean primarily the opinions of Professor E.C. Tolman (19), and secondarily those of his various students all of whom have criticized the S-R hypothesis and have emphasized the importance of a new approach to the analysis of the learning process (13).

to state the S-R hypothesis in as precise a fashion as possible.

Now whenever one engages in such a task as this, trying to increase the precision of some statement which is vague, one inevitably faces two criticisms. Some critics will make the accusation that the increase in precision has resulted in such a narrow statement that much which was connoted by the original vague statement is not included. To such critics my only reply is that the original hypothesis was vague because it contained many conflicting or inconsistent connotations, and my purpose in restating it more precisely is to remove all these connotations which I believe are irrelevant. Other critics will make the contrary accusation. They will claim that I have said no more than was contained in the original statement. To these critics I would also reply that my intention was to say no more than was contained in the original S-R hypothesis, but to say it more precisely, and to say it in a way that would enable us to determine whether or not the hypothesis is true.

2. Dispositions and Conditioned Definitions

Before I can present my analysis of the S-R hypothesis, I must explain what I mean by a "disposition" and by a "conditioned definition", as distinguished from an ex-

plicit definition.

Consider for a moment the following properties of objects--solubility, fragility, elasticity, permeability and so forth. Now when we say that this piece of sugar is soluble in water, we do not mean that it is now dissolved; we mean, rather, that this piece of sugar has a disposition to dissolve when put in water. Now suppose that we wish to study and talk about the class of things which have this disposition, because we believe that they may have other properties in common which further distinguish them from the things which do not have this disposition. It will be convenient in this case to have a name for this class since we will be referring to it frequently, and in order to give this class a name we must introduce a definition.

But before I try to write such a definition, let us examine the rules which seem to govern customary usage. That is, let us consider the conditions under which such usage decides that the sentence "A has the disposition to dissolve in water" is true, false, and the conditions under which no decision can be made. It will be generally agreed that according to customary usage, this sentence is (1) considered <u>true</u>, when A is in water and A dissolves; (2) considered <u>false</u>, when A is in water, and A does not dissolve; and (3) no decision can be made about its truth

when A is not in water.

The problem is now to discover a logical form which we can use to introduce the matrix[3] "x has a disposition to dissolve in water". The molecular form[4] that we require is one which will accord with the truth values of customary usage, which we have just described.

Carnap (5) has demonstrated that disposition predicates cannot be introduced by means of an explicit definition like the following one

1.0 x has the disposition D $\equiv$ x is in water $\supset$ x dissolves.[5]

without serious logical difficulties. The main difficulty is that according to such a definition, "A has the disposition D" will be true whenever "A is in water" is false. As a consequence we must reject such an explicit definition and employ some other molecular form. Carnap (5) suggests the following form which he calls a "bilateral reduction sentence" :

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3. A matrix is an expression which contains a free variable. When a value is specified for this variable, and the name of this value is substituted for the variable, the matrix becomes a sentence.

4. A sentence has a molecular form when it is composed of two or more simple or atomic sentences joined by logical connectives like "and", "either or", "implies", or "equivalent to".

5. For the sake of convenient recognition and reference we will designate all definitions with numbers in the range from 1.0 to 9.9, while all synthetic sentences or laws will be given numbers in the range from 10.0 to 19.9. For explanation of the logical symbolism employed throughout see Appendix II at the end of this paper.

2.0 $P(x) \supset (Q(x) \equiv R(x))$

where R(x) is the matrix to be introduced (ie., "x has the disposition D") and Q(x) and P(x) are matrices which have been defined elsewhere. When we employ this form for introducing new expressions into the language we call the entire expression a "conditioned definition". Now let us consider such a conditioned definition for the matrix "x has a disposition to dissolve in water".

2.1 x is in water $\supset$ (x dissolves $\equiv$ x has a disposition to dissolve in water)

From the truth-table of such an expression we can discover that when P(x) is true, the truth value of R(x) is identical with that of Q(x), (ie., if Q(x) is false, R(x) is false; if Q(x) is true, R(x) is true). On the other hand when P(x) is false, the relation between Q(x) and R(x) is undetermined (ie., Q(x) and R(x) may have identical truth values and they may not). Thus we see that the truth values of P(x), Q(x), and R(x) in this molecular form conform precisely with those in customary usage. Now we may use the matrix "x is soluble in water" as an abbreviation for the longer expression, by writing the following explicit definition,

2.12 x is soluble in water $\equiv$ x has a disposition to dissolve in water.

Now it should be pointed out that there are two kinds

of dispositions, those which are relatively permanent, and those which are transitory. (1) Thus, if a certain thing is soluble in water in one set of conditions, we may discover that it is also soluble in many other circumstances. In this sense solubility is a relatively permanent property of a substance, and might be considered as an essential rather than an accidental property. (2) Volatility on the other hand is a less permanent kind of disposition. Thus, we discover that some liquids are volatile under certain conditions (eg. when they are hot, and when the pressure of the surrounding atmosphere is low, etc.) but not volatile under other conditions. Let us consider for a moment how this disposition predicate is employed in customary usage, and then how the matrix "x is volatile" should be defined.

According to customary usage we say that "A has a disposition to evaporate when exposed to air" is (1) considered <u>true</u>, when A is exposed to air, and A evaporates; (2) considered <u>false</u>, when A is exposed to air, and A does not evaporate; and (3) that no decision can be made when A is not exposed to air. From this analysis it is clear that the following conditioned definition will conform to this usage :

2.2 x is exposed to air $\supset$ (x evaporates $\equiv$ x has a disposition to evaporate when exposed to air)

and we may then write the following explicit definition :

2.21 x is volatile ≡ x has a disposition to evaporate when exposed to air)

Now as I mentioned above we know, that is we have discovered, that whether a liquid is volatile or not depends upon its temperature and the pressure of the surrounding air. Once we have defined the matrix "x is volatile" as we did in 2.21 we can express this discovery by the following law:

10.0 For all x, x is hot and the pressure of the air surrounding x is low ≡ x is volatile.[6]

This law is a synthetic sentence and its truth depends not upon logic but upon the facts which we have discovered. The law is true only if both of the following implications are true : (1) if any of the specified liquids is hot, and the pressure is low, then this liquid is volatile and (2) if any of the specified liquids is volatile, then this liquid is hot and the pressure surrounding it is low.

Now with these technical notions made clear we can turn to a consideration of the facts of classical conditioning which gave rise to the S-R hypothesis.

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6. The range of the variable in this law is a specific class of liquids.

3. The Definitions and Laws of Classical Conditioning

It is generally recognized that the S-R hypothesis is today used primarily to describe the changes in behavior which result from a particular kind of training procedure which has been called by Hilgard and Marquis (7) "the classical conditioning procedure". They describe the three essential features of this procedure in the following way :[7]

> (1) An unconditioned stimulus (eg. an airpuff to the eye) which, in the experimental situation evokes a regular and measurable unconditioned response (eg. an eyelid response), (2) a conditioned stimulus (eg. a light) which originally does not evoke the unconditioned response, and (3) repeated presentation of the conditioned and unconditioned stimuli in a controlled and specified way." (7)

Now when a subject has received a certain number of repetitions of these stimuli, his behavior changes and he begins to make the eyelid response before the occurrence of the airpuff.

This change in behavior has usually been described as one in which a new stimulus-response connection or bond is formed. What is emphasized is that, although the eyelid response which is connected with the light may differ in

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7. In this and the following quotations I have chosen the airpuff, the light, and the eyelid response for examples. Thus I have made the appropriate changes in the quoted material in order to keep the examples the same. I have chosen the eyelid response because it is generally agreed to be one of the most dependable ones with adults and also animal subjects. (Vide Woodworth (25)) It was also important for clarity to choose an example in which motivational states of the subject would be relatively unimportant.

some ways from the original unconditioned eyelid response connected with the airpuff, the relation which connects both responses to their respective stimuli is essentially the same. Thus Hilgard and Marquis (7) describe the behavioral change by saying that

> a relationship has been established between (the light and the airpuff), so that a response originally made to (the airpuff) is now made to (the light). (7)

Liddell describes the change in a slightly different way by saying that the light becomes

> a signal for (airpuff) and evokes (the eyelid response). (The light) which previously had no effect upon (the eyelid muscles) has now acquired the same property of stimulating (the eyelid response) as (the airpuff). (10)

While Hull says that such training results in setting up

> an increment to the associated connection between the different receptor impulses produced by (the light) and (the eyelid response). Thus after a sufficient number of repetitions there must arise the new super-threshold receptor-effector connection, S_c ---- R_u. (8)

Let us examine the notion of "connection". The Oxford Concise Dictionary gives the following three meanings for the verb "to connect" : 1. to join two things one with another; 2. to be related to; 3. to associate mentally. Now to say, according to meaning 1, that a stimulus is joined

or fastened to a response, is to say something obviously false. To say, according to meaning 2, that a stimulus is related to a response, is to say something essentially vague, since the nature of the relation remains unspecified. And, finally, to say that a stimulus is mentally associated with a response, is to say something meaningless until we know what is meant by the expression "mental association".

But the reader may now object that if I am asserting that all S-R psychologists have talked nothing but nonsense, I have overlooked the fact that they are able to communicate well enough to produce many experimental facts upon which most of them agree. Since such communication and agreement would be impossible if their notion of "stimulus-response connection" were meaningless, they must, the reader concludes, have meant something by this expression, and further, they must all have meant much the same thing. My reply to this is two-fold. First of all, I do not believe that the notion of "stimulus-response connection" is meaningless, but only that it is very vague. And, secondly, it should be clear that where there is vagueness agreement is easy, for as Bertrand Russell has pointed out, "One can almost always hit the bull's eye with a piece of putty."

Can we conclude then that so far no one has given a clear and precise meaning for this expression? Since I know of no such account, I will henceforth assume that none

has been made. Whether or not the task is worthy trying is, of course, another question, and to some the following analysis may appear to be cracking peanuts with a sledge hammer. However, the only excuse I can give for undertaking this analysis is that it will make clearer the differences between the opinions of the Tolman school and those of the S-R school, which is the pain purpose of this introduction.

So let us begin again. Consider a subject who has received a series of trials in which a light directed toward his eye at time t has been followed at time t + e by an airpuff directed to the same eye. How shall we describe the changes in the subject's behavior which result from this training? One possibility is to say that the subject has acquired a disposition to make an eyelid response when a light is presented to his eye. Let us examine the following conditioned definition for this disposition

3.0 x is presented with a light $\supset$ (x makes eyelid response $\equiv$ x has a disposition to make an eyelid response when a light is presented).

Now there are several reasons why this definition is not adequate. Definition 3.0 would not enable us to distinguish "chance" eyelid responses to the light from those that are habitual. We have discovered that both kinds of

responses occur in our subjects, and we wish to talk only about the disposition to make habitual responses. One way of making this distinction might be to include in our definition some statement about the fact that over a long series of trials the subject makes the eyelid response whenever the light is presented. But to try to include this statement in our definition would probably raise logical difficulties which we could not remove. However, since there is a much simpler way of achieving the same result, I will choose it. This method is to specify in the definition the time relation between the occurence of the light and the response. We know that when a subject has a disposition to make these responses habitually, the latency of the eyelid response remains relatively constant, and precedes the time at which the puff has been presented during the training series. Now if we call the time at which the light begins "t" and the time at which the puff has been presented "t + e" we may then specify in the definition that the eyelid response occurs before t + e. Such a further specification in definition 3.0 should greatly reduce the difficulty in distinguishing "chance" from habitual responses.[8] But not only should we specify

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8. Of course this method does not completely guarantee that we shall be able to distinguish chance from habitual response, since chance responses may still occur before t + e. But no criterion we might choose would be any better a guarantee.

that the eyelid response occurs before t + e, but we ought also to specify that it occurs after t + θ, which is the latency of the natural eyelid response to light. Thus we may now write

3.1 x is presented with light at t ⊃ (x makes eyelid response after t + θ and before t + e ≡ x has a disposition to make an eyelid response when light is presented)

Definition 3.1 satisfies the empirical requirements discussed so far, but it does not satisfy another very important one. Suppose that an airpuff[9] were presented between t and t + e. We would then be unable to decide, when the subject made the eyelid response, whether or not he had the disposition. Therefore we should include in the antecedent of the conditioned definition, the statement that no airpuff is presented between t and t + e. We now write

3.12 x is presented with light at t. No airpuff is presented between t and t + e ⊃ (x makes an eyelid response after t + θ, and before t + e ≡ x has a disposition to make an eyelid response when a light is presented and no airpuff is presented)

---

9. There are several reliable natural stimuli for the eyelid response besides an airpuff, such as shock, a blow on the cheek, a sharp sound, etc. For the sake of simplicity, however, we will consider only the airpuff, since what is true of it applies also to the other stimuli. Thus the reader is asked to consider "airpuff" as equivalent to "airpuff or shock or blow on the cheek or a sharp sound."

Now one further point remains to be considered. It is a well established fact that even though a subject may have a disposition to blink when the light is presented, if a distracting stimulus is presented after the presentation of the light, the subject will not respond. Do we wish to say that under these conditions the subject does not have the disposition? If we adopt definition 3.12, we shall be forced to make this denial. Or do we wish to say that under these circumstances we cannot decide whether or not he has the disposition? If we decide in this latter way, we shall have to abandon definition 3.12. Our decision in this matter should be based upon some belief about which of the two different definitions and their respective dispositions are likely to be the more fruitful for the purposes of investigation. My belief is that we should abandon definition 3.12 and choose the second alternative. To do this we must include in the antecedent of the conditioned definition, a statement that there is no distracting stimulus between t and t + e. Thus I present the following conditioned definition as one which meets all the logical and empirical requirements which we have discussed

3.2 x is presented with light at t. No airpuff and no distracting stimulus is presented between t and t + e $\supset$ (x makes an eyelid response after t + θ, and before t + e $\equiv$ x has a disposition to make an eyelid response when a light is presented and no airpuff or distracting stimulus is presented.

Now we may for the sake of convenience choose an abbreviation for the long expression "x has a disposition etc." and introduce this appreviation by the following explicit definition :

3.21 x has a conditioned eyelid response habit to light $\equiv$ x has a disposition to make an eyelid response when a light is presented, and no airpuff or distracting stimulus is presented.

This analysis began with the purpose of finding some precise way of describing the changes in a subject's behavior which result from a particular kind of training. We rejected all descriptions of this change as a change in S-R connections because they were too vague. But if we now adopt definition 3.21 we can describe the changes in behavior very precisely by stating that a subject, who has received a particular kind of training, has acquired a conditioned eyelid response habit to light. This, of course, is a synthetic sentence, and thus can be expressed as a law if the training procedure is described precisely. The following then might be the first law of classical conditioning :

11.0 For all x and all t, x is given N trials in each of which a light is presented at t and an airpuff at t + e $\equiv$ x has a conditioned eyelid response habit to light.

As I pointed out in the preceding section a law which has this form (ie. (x) (P(x) $\equiv$ Q(x)) is true only if two impli-

cations are true, namely if (x) $P(x) \supset Q(x)$ is true and if (x) $Q(x) \supset P(x)$ is true. Thus law 11.0 is true only if both of the following statements are true : (1) if any subject is given this training, then this subject has a conditioned response habit of this kind and (2) if any subject has a conditioned response habit of this kind, this subject must have had this training.

Let us now consider the evidence for 11.0. There seems to be no evidence against the statement that if a subject is given this kind of training, then he will acquire a conditioned response habit of this kind. Thus the first implication of law 11.0 may be true; at least, we have no evidence that it is false. The critical reader may comment at this point that this statement is only true when the laboratory conditions are controlled in a specific way, and when the subject is prepared in a specific way for the experiment. This is, of course, true. But, when these specific conditions are included in the description of the training procedure, there seems to be no evidence at present which contradicts the statement that this kind of training produces a conditioned response habit.[10]

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10. It should be emphasized that the training procedure specified in law 11.0 is one in which there is a constant temporal relation between the airpuff and the light. This procedure should not be confused with one in which the airpuff follows the light only if the subject fails to make a response. Hilgard and Marquis (7) call this latter procedure "avoidance training". Thus evidence drawn from experiments involving avoidance training have no relevance to the first implication of law 11.0. We shall show below that such evidence is relevant, however, to the second implication

On the other hand there is a great deal of evidence which contradicts the second implication of 11.0, that if any subject has a conditioned response habit of this kind, this subject must have had this training. First of all the facts of higher order conditioning indicate that there are some subjects who have a conditioned eyelid response to light, but who have not been trained with light and airpuff (or shock or sound etc.). Instead such subjects have first been trained with, say, a bell and an airpuff, and then later trained with light followed by a bell and no airpuff.

Secondly, the facts of avoidance conditioning also contradict the second implication of 11.0, since there are subjects with the conditioned response habit, but who have not been trained with light and airpuff in a constant temporal relation.[11]

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11. There is considerable verbal difficulty in distinguishing these two training procedures since in both procedures whenever the airpuff (or shock etc.) occurs, it occurs at a specific interval after the light. But the difference between the two procedures can be stated precisely in the following way. Thus let "P" mean "x is presented with light at t"; "Q" mean "x is not shocked before t + e"; "R" mean "x does not respond before t + e"; and "S" mean "x is shocked at t + e". Then avoidance training may be described in the following way :

    For all t, $P(x).Q(x).(R(x) \equiv S(x))$

In contrast to this complex expression, classical conditioning procedure may be described in the following simple fashion :

    For all t, $P(x).S(x)$

Instead, such subjects have been given some trials in which the light at t was not followed by an airpuff at t + e. These were the trials in which the subjects made an eyelid response before t + e.

Finally one other kind of experiment provides further evidence against the second implication of 11.0. Brogden (3) presented a light followed by a bell to a dog for 200 trials. He next taught the dog to retract its leg to the bell, by following the bell with shock. Then, when he presented the light again, although it had never been followed by shock, the dog retracted its leg. At first glance his experiment appears to be another case of higher-order conditioning, but it should be remembered that the light and bell were paired, before the bell had been paired with shock.

Thus we have discovered four distinct kinds of training procedures which will produce conditioned response habits.[12] At the present stage of inquiry we may now choose

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12. The reader may have wondered why we have not mentioned evidence drawn from reward training experiments. The reason for this omission is that such evidence is irrelevant to law 11.0. Why it is irrelevant will become clear when definition 3.2 and 3.21 are considered. Reward training does not produce the disposition defined in 3.2, but produces instead a disposition to make a response when the subject is hungry (or deprived). Since this condition is not included in the antecedent of definition 3.2, evidence drawn from reward training experiments is irrelevant to law 11.0.

between two alternative ways of stating these facts. (1) If we let "P" name the classical or Pavlovian procedure, "H" name the higher-order procedure, "A" name the avoidance procedure, and "B" name Brogden's procedure, we can then write four separate laws : (x) $P(x) \supset Q(x)$; (x) $H(x) \supset Q(x)$, (x) $A(x) \supset Q(x)$, and (x) $B(x) \supset Q(x)$. Or (2) we can include all four of these procedures in a stating of an equivalence in the following way :

11.1 $(x)\ P(x) \vee H(x) \vee A(x) \vee B(x) \equiv Q(x)$

which may be read

> For all x, x is given training P or H or A or B $\equiv$ x has a conditioned eyelid response habit to light.

The main theoretical problem that then faces us is to find some characteristic or set of characteristics which all four procedures have in common. Should we be able to discover such common characteristics we would then be able to write the law

$$(x)\ S(x) \equiv Q(x)$$

Where S(x) is an abstract description of a training procedure which includes P, H, A and B as various sub-types of this general proceudre. It should be noted that care must, of course, be taken to avoid stating S(x) is such a way that it also includes some other procedure, say C, which does not produce a conditioned eyelid response habit.

It is a law of this general kind which most of the

traditional theories of learning were trying to formulate. The principle of substitution, and the principle of effect, thus may be viewed as alternative hypotheses about the essential features common to all the various training procedures which produce conditioned response habits. Many c criticisms have been made of both principles, some of which are relevant and some of which are not. It should be clear, now, however, that neither of the principles as they have usually been stated, can replace the expression "P(x) v H(x) v A(x) v B(x)" in law 11.1 and the law remain true. It would not be difficult to prove this last statement, but since it would be a digression from my main purpose, I will not present the proof. Instead, I will turn to a discussion of an important empirical requirement which must be fulfilled by a conditioned definition. The reason for considering this requirement is that many definitions in the literature of psychology seem to violate it, and as a consequence much confusion has resulted. To avoid similar confusions we must consider this requirement carefully before we consider the S-R hypothesis in its relation to spatial learning.

4. The Empirical Requirement of a Conditioned Definition

So far I have illustrated what I believe is the correct way to state the basic facts in classical condition-

ing. In doing this I took pains to indicate some of the empirical and logical problems that must be resolved if we are to talk precisely. For example, I developed the conditioned definition for the matrix "x has a conditioned eyelid response habit to light" through several stages from definition 3.0 to definition 3.21, explaining at each step why we must revise the previously suggested definition. After this I presented what I called "the first law of conditioning" and considered the experimental evidence which is relevant to this law.

Now many of the definitions in the literature of psychology, especially those which are called "operational", appear to be quite similar to mine. As a consequence some readers may have wondered how my definitions differ from the so-called operational ones. In answering this question, I will confine myself to emphasizing only one point.[13] There is an important empirical requirement which all conditioned definitions must satisfy. Since many of the so-called operational definitions do not seem to satisfy this requirement, I will consider this to be the essential dif-

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13. There is so much verbal confusion among psychologists about what constitutes an operational definition that it would be useless to try to clarify what they might mean, and then to compare this with what I mean by a conditioned definition. To the reader who doubts that the psychologists are confused I suggest the "Symposium on Operationism" (2)

ference between the two kinds of definition.

Let us consider the molecular form $P(x) \supset (Q(x) \equiv R(x))$ when R(x) is the matrix which contains the expression which is being introduced and P(x) and Q(x) contain expressions which have been defined elsewhere. Now the empirical requirement which all definitions of this form must satisfy is that P(x) and Q(x) have no empirical relation. Let me explain the reason for this requirement. Let us suppose that we have discovered the following law to be true.

$$(x)\ P(x) \supset Q(x)$$

In this case P(x) and Q(x) are empirically related in such a way that whenever P(x) is true, we find that Q(x) is also true. Now remember that in any conditioned definition the truth value of $Q(x) \equiv R(x)$ is undetermined except when the antecedent, P(x), is true. That is, when P(x) is false we cannot determine the truth value of R(x) from the truth value of Q(x). But in our assumed law we have discovered an empirical relation between P(x) and Q(x) such that whenever P(x) is true, Q(x) is also true. This means that whenever the conditions, P(x), which enable us to determine the truth value of R(x) from Q(x) are satisfied, Q(x) is always true. As a consequence it will be impossible to find a case where R(x) is false, if the law $(x)\ P(x) \supset Q(x)$ is true.

An example may make this point clearer. For the pur-

poses of this example let us suppose that we have discovered that if a subject has been given training B and is now presented with a light, then the subject makes an eyelid response. Now let us further suppose that we define the matrix "x has a conditioned eyelid response habit to light" in the following way.

4.0 $\hat{x}$ x is given training B and is now presented with light $\supset$ (x makes an eyelid response $\equiv$ x has a conditioned eyelid response habit to light)

Can we, according to definition 4.0, distinguish between those subject who have conditioned eyelid response habits and those who do not? We cannot. According to definition 4.0 we can only assert that the given subject does not have a conditioned response habit when the antecedent of definition 4.0 is true and the second matrix of 4.0, "x makes an eyelid response", is false. But since $(x)\ P(x) \supset Q(x)$ is assumed to be empirically true, there will be no cases when P(x) is true and Q(x) is false. As a consequence we will not be able to distinguish subjects who have the habit from those who do not if the first and second matrices of our conditioned definition are empirically related. Since the original purpose of introducing the definition was to enable us to make this distinction, any definition which violates this empirical requirement will be useless.

This point is, I believe, an extremely important one for psychologists to remember, since many of the definitions

which they have proposed seem to violate this requirement. For example, consider the following proposed definition for a conditioned response habit

> The process of conditioning is distinguished by what is done to the organism to induce the change; in other words, it is defined by the operation of simultaneous presentation of the reinforcing stimulus and another stimulus. (15)

Or take the following example drawn from the glossary of Hilgard and Marquis (7)

> Conditioned Response - A response which appears or is modified as a consequence of the occurrence of a conditioned stimulus in proximity to reinforcement. (7)

Although neither of these definitions are stated in a precise form, they both appear to violate the empirical criterion. Such examples could be multiplied indefinitely.

Now the tendency to write definitions of this kind arises, I believe, from a basic confusion which is common to many psychologists who have been exposed to logical positivism and operationalism. This confusion results from a failure to distinguish (1) the conditions, $P(x)$, which we have decided must be fulfilled if $Q(x) \equiv R(x)$ is to be considered true, from (2) the conditions we have discovered are fulfilled when $Q(x)$ is true. Thus in the definitions we have chosen for example, there is a confusion between (1) the conditions (ie, the presentation of the light, etc.) which must be fulfilled if we are to take the occurrence

of the response as evidence that the animal has a certain habit[14] and (2) the conditions (ie. the training procedure, etc.) which we have discovered are sufficient if an animal is to make this response.

5. What Is An S-R Connection?

So far I have presented an analysis of what is, or, at least, of what should be meant by the expression "x has a conditioned eyelid response habit to light". It has become clear in this analysis that to formulate precise and useful definitions and laws, many tiresome details must be specified. Since most psychologists have not troubled themselves with making such an analysis, but instead have taken the expression "S-R connection" as unanalyzed, these details have generally been ignored. Rather than spend their time in clarifying the meaning of this expression they have preferred (1) to employ this new experimental technique in a wide variety of new situations (eg. with different kinds of subjects, and different kinds of stimuli, etc.) and (2)

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14. This is admittedly a loose way of talking. When I say that the occurrence of the response, $Q(x)$, is taken as evidence that the animal has the habit, $R(x)$, I do not mean, of course, that we infer the truth of $R(x)$ from $Q(x)$. An inference can be made only when two predicates (eg. $S(x)$ and $T(x)$) are defined independently, we have discovered a law relating them (eg. $(x)\ S(x) \supset T(x)$) and we now observe $S(x)$ to be true. In this latter case, we can infer on the basis of the law the truth of $T(x)$ from $S(x)$. But when I say that we take the response as evidence that the animal has the habit, all I mean is that under these conditions, $P(x)$, the matrix "x makes the response" means the same thing as "x has the habit".

to use the notion of S-R connection to explain or interpret the behavior of animals in other kinds of learning situations, that is, where the paired stimuli have not been controlled by the experimenter.

Thus, during the nineteen twenties when behaviorism was most aggressive, every change in behavior was interpreted as an instance of conditioning. From the training situation two stimuli were abstracted, one of which was called "the conditioned" and the other the "unconditioned stimulus", while the new behavior was called "the conditioned response". The change in behavior, described as a new "S-R connection" was interpreted either by a neurological analogy, or by S-R diagrams.

During this early period the neurological analogy was the more popular interpretation. The following passage from Skinner makes clear the motives which made this interpretation so appealing.

> We have been proceeding, of course, upon an unnecessary assumption, namely, that there is a flexion reflex which exists independently of our observations, and which our observations approximate. Such an assumption is wholly gratuitous, but it is remarkably insistent.... If we remain at the level of our observations, we must recognize a reflex as a correlation. But the immediate uncritical reaction to a definition on that basis is that a correlation, in point of satisfaction, is not enough. There is an urge toward solidification, clearly evident throughout the history. We turn instantly to the reflex arc for material support. (14)

To me this passage indicates that the failure to clarify the notion of "S-R connection" beyond stating vaguely that it consists of a correlation, resulted in a feeling of insecurity. Such feelings often arise, or atleast should arise, when we recognize that we do not know what we are talking about. Since the notion of the reflex arc was, anatomically at least, a precise one, the psychologists felt it was better to talk about that, even if it was irrelevant, than not to know what they were talking about. Today it is widely recognized that this appeal to the reflex arc was irrelevant.[15] As long as the notions we use to describe the changes in gross behavior are vague and ambiguous; no neurological facts can possibly help to explain these changes. Only when we have been able to state the laws of conditioning precisely, will it be possible to relate such laws to the laws of neurology. At present there are no laws of either kind, and so we may conclude that the neurological analogy was simply a device for evading the problem of clarifying the notion of "S-R connection".

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15. Hilgard and Marquis state flatly : "The known facts of neural function cannot be utilized to predict or to limit the results of behavioral studies.... Even the basic law that a response varies with the intensity of the stimulus would be equally true if the nerves were copper wires or pneumatic tubes. The facts of speed of conduction and synaptic delay cannot predict the latency of a conditioned response, for we have no idea what length of nerve or how many synapses are involved... Many of the so-called neural facts, such as reflex inhibition, which seem most relevant to conditioning are in reality behavioral laws stated as relations between afferent and efferent nerve activity without direct observation of any intermediate neural events." (7)

The S-R diagram was the other device for evading this problem. This device was even more popular since the precision with which the digrams were drawn disguised the vagueness of the notions which they were intended to illustrate. In fact, the deception was so effective that even the opponents of S-R psychology, such as Lewin and Tolman, took up diagrams to represent their notions. But it should be clear that such precision was specious. Only when we are clear about the relations which we wish to diagram, can we be sure that the diagrams we draw represent these relations accurately.[16]

Recently a new device for disguising the vagueness of the notion of "S-R connection" has come into style. The following quotation from Spence (16) indicates that this disguise was contrived as a result of dissatisfaction with the neurological analogy.

> The stimulation of an end organ by the conditioned stimulus (Sc) is assumed to initiate a central neural activity or stimulus trace.

Then in a foot-note Spence continues:

> While referred to as a form of neural activity, it should be clearly understood that this construct of stimulus trace is not operationally defined in terms of the instruments and experimental operations of the neurophysiologist. It is defined, by mathematical equation, as a function of experimental variables which the psychologist himself

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16. Thus models of atomic structures would be meaningless if it were not possible for us to describe the relations without recourse to the diagrams.

> works with, that is, intensity of the stimulus (S), duration of the action of S since its beginning (T), and the time since the termination of the stimulus (T'). The mathematical equation defining stimulus trace (s) is, then,
>
> $$s = \underline{a} \log S (1 - \underline{e}^{-hT}) \underline{e}^{-kT'}),$$
>
> where $\underline{a}$, h, and k are empirical constants, with the specification that $h > k$. (16)

I find this passage totally confusing. The source of the confusion seems to lie in the failure to distinguish the problem of defining a matrix like "x is a stimulus trace", from the problem of measuring some property (eg. the intensity) of a thing or process which we have already named "a stimulus trace". When we distinguish these two problems we should recognize immediately that it is absurd to say that we will define "stimulus trace" by a mathematical equation. The variables in a mathematical equation take numbers as their values. Thus to say that "stimulus trace", or "habit", or "reaction potential", etc. are defined by mathematical equations, is to say that if there are any such things (eg. stimulus traces, etc.), they are numbers. Only Pythagoras would defend such an extreme position!

In this section I have described three alternative kinds of answers which psychologists have given to the question "What is an S-R connection?". In discussing these three answers I have argued that all three of them are devices for disguising the fact that the notion of "S-R connection" is fundamentally vague.

I want now to discuss the problems involved in extending S-R psychology to other kinds of learning situations (eg. maze learning, discrimination learning and problem solving). Since the S-R psychologists have not troubled to analyze their notions, I will presume that my analysis of the conditioned eyelid response habit conforms to their notions as well as any relatively precise notion can conform to a vague one. With this presumption we may now turn to a consideration of the problems arising from the extension of S-R psychology to other kinds of learning.

6. Two Assumptions Implicit in S-R Psychology.

There are two important assumptions implicit in any extension of S-R psychology. The first assumption is that some given principle of reinforcement states the essential factors common to all the training procedures which produce habits. The second assumption is that all habits are essentially similar. Both of these assumptions are necessary if other forms of learning are to be explained in terms of the principles of conditioning. The reason why this is the case is easily explained. All such explanations are based upon the inference that if an animal has a given habit, then its training or past experience must have been of a certain kind. But such an inference is justified only if the two assumptions I am talking about are true. Let us then examine these assumptions to see if they are true. I will

begin with the first one.

After describing several different simple learning experiments, Hull, in his Principles of Behavior (8) comes to the following conclusion :

> An inductive comparison of these superficially rather divergent forms of learning shows one common principle running through them all. This we shall call the law of primary reinforcement. It is as follows : Whenever an effector activity occurs in temporal contiguity with an afferent impulse, or the perseverative trace of such an impulse, resulting from the impact of a stimulus energy from a receptor, and this conjunction is closely associated in time with the diminution in the receptor discharge characteristic of a need, there will result an increment to the tendency for that stimulus on subsequent occasions to evoke that reaction. (8)

And to account for the process in which habits are unlearned, Hull cites the following principle, which is the converse of the previous one.

> When a reaction is evoked repeatedly without a closely associated drive reduction, the power of the stimulus-motivated combination to evoke the response in question gradually diminishes. (8)

From these remarks we may conclude that Hull believes that his principle of primary reinforcement states the essential factors common to all training procedures which produce habits. Now let us examine the evidence for this assumption when it is restricted to the kinds of training procedures which produce a conditioned eyelid response habit. The question is this. Can the principle of primary reinforcement be substituted for the expression "P(x) v H(x) v A(x) v B(x)" in law 11.1, and the law still remain true?

Now let us consider each of the four training procedures to determine whether all of them conform with Hull's principle. If any one of them does not so conform, then his principle does not account for the conditioned eyelid response habit.

In procedure P, the light presented at time t is followed by an airpuff at t + e. The procedure consists in giving repeated presentation of this pair of stimuli in this temporal order. In this case the "effector activity" (of the blink) occurs in temporal contiguity with the "afferent impulse" (from the light). The primary question, then, with procedure P is whether or not the occurrence of the response is "associated in time with diminution in the receptor discharge characteristic of a need." There seems to be no good reason for supposing that this requirement is satisfied by procedure P. But some readers might insist here that the cessation of the airpuff constitutes the reduction of a need. This argument, however, assumes that the onset of the airpuff causes a state of drive or motivation. Therefore the argument is invalid since the learned blink is closer temporally to an increase in drive than it is to drive reduction.

The same considerations apply pari passu to procedure A, avoidance training. But one difference between P and A ought to be mentioned. After the subject has acquired

a conditioned eyelid response habit by procedure A, the response is no longer followed by an airpuff. Thus, no matter how one may interpret the role of the airpuff, it certainly plays no role in maintaining the habit after the habit has been acquired.

While Hull recognizes that procedure H, higher-order conditioning, does not conform with his principle, he does not discuss Brogden's procedure.[16] But in both cases it should be obvious that Hull's principle is irrelevant, and that Hull has completely failed to state a principle of reinforcement that can be substituted in law 11.1. Let us now turn to the second and theoretically more important assumption implicit in S-R psychology.

To extend the laws of conditioning to other kinds of learning situations it is not only necessary to state an adequate principle of primary reinforcement, but it is also necessary to prove that the dispositions acquired in the more complex learning situations are the same as the dispositions acquired by classical conditioning. In short, one must show some evidence that all acquired dispositions are essentially similar. Advocates of the S-R psychology

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16. Hull not only recognizes that the higher-order procedure does not conform with his principle, but says that this procedure yields a new principle, secondary reinforcement, which he believes explains all other procedures which do not conform to the principle of primary reinforcement. This argument, of course, begs the question as is pointed out in an article by the present writer called "Hull's Treatment of Learning". (13)

have all assumed that this is true. For them all learning consists in the acquisition or strengthening of "S-R connections". This dogma is so widely accepted that few psychologists have recognized that it is based upon an empirical assumption for which evidence is needed.

Since this assumption asserts that all acquired dispositions are essentially similar, it follows that all the dispositions acquired by the various classical conditioning procedures must be the same.[17] Let us see whether this is true.

Consider Pavlov's original experiment with conditioned salivation. Hungry dogs were presented with a tuning fork at t and given a small amount of dry food at t + e. After a certain number of repetitions of this sequence, the dogs developed a disposition to salivate to the tuning fork alone. Now is this conditioned salivary response habit the same kind of disposition as the conditioned eyelid response habit, introduced by definition 3.2? This question can be answered by examining the definitions for the two dispositions. If the conditioned definition for the salivary

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17. By "classical conditioning procedures" I mean all those procedures in which paired stimuli are controlled by the experimenter in such a way that the temporal relation between these stimuli remains fixed. Thus "classical conditioning" includes procedures P, A, H, B, and perhaps others, but excludes maze and discrimination learning, and other forms of complex learning.

habit is exactly like the one for the eyelid habit, then the two dispositions are similar.[18] Otherwise they are not. The following definition is exactly like 3.2 except that I have, for the sake of simplicity, substituted "x has a conditioned salivary response habit to a tuning fork" for the long expression "x has a disposition etc."

5.0 x is presented with tuning fork at t and no food and no disturbing stimulus is presented between t and $t + e \supset$ (x salivates after $t + \theta$, and before $t + e \equiv$ x has a conditioned salivary response habit to a tuning fork).

Now the disposition defined by definition 5.0 is exactly like the disposition defined by definition 3.2 for the conditioned eyelid response habit. But is this disposition the one we wish to concern ourselves with? Is it the one which has traditionally been meant by the expression "a conditioned salivary response"? I do not believe that it is. Let me explain.

Suppose we have trained a dog as Pavlov did, and we now wish to determine whether or not the dog has the conditioned salivary response habit to a tuning fork. Suppose further that the dog is satiated at the time of this test. Now when we present the dog with the tuning fork at t, and no food and no distracting stimulus between t and t + e, the dog does not salivate before t + e. Do we

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18. The two conditioned definitions will be exactly alike when we merely substitute "tuning fork" for "light", "no food" for "no airpuff" and "x salivates" for "x makes eyelid response".

wish to say under these circumstances that the dog does not have the conditioned salivary response habit? If we adopt definition 5.0 we shall be required to say this. Furthermore this latter assertion would entail that the following law is false :

12.0 For all x and all t, x is given N trials, in each of which x is deprived of food and a tuning fork is presented at t and dry food at t + e ⊃ x has a conditioned salivary response habit to a tuning fork.

The problem here is exactly like the one of external inhibition which we discussed on page 15. There we solved the problem by including the matrix "x is not presented with a distracting stimulus between t and t + e" in the antecedent of the conditioned definition. This solved the problem by making it impossible to determine whether or not a given subject had the disposition when a distracting stimulus is presented. The disposition we are concerned with now is a disposition to salivate when deprived of food, presented with a tuning fork, etc. Thus, although it has never been formally stated, it has been customary to use this notion in such a way that it is impossible to determine whether or not a given subject has the habit when the subject is satiated. This means that if we are to save law 12.0, we must conform with this customary usage, and include "x is deprived of food for N hours before t" in the antecedent of the conditioned definition. Thus we may now

write

5.1 x is deprived of food for N hours before t and x is presented with a tuning fork at t and no food and no distracting stimulus is presented between t and $t + e \supset$ (x salivates after $t + \theta$ and before $t + e \equiv$ x has a conditioned salivary response habit to a tuning fork).

In conclusion, then, if we are to save law 12.0, we must adopt a definition which defines a disposition quite different from the eyelid disposition. To one who wishes to maintain the second assumption of S-R psychology, namely that all dispositions produced by classical conditioning are the same, and who wishes also to maintain law 12.0, the dilemma is inescapable. Either law 12.0 is false or the conditioned salivary response habit is not the same kind of disposition as the conditioned eyelid response habit.[19]

Thus the second assumption implicit in S-R psychology is false. The dispositions acquired in all learning situations are not the same. This is, I believe, an extremely important point and it constitutes the main theoretical difference between the S-R psychologists and those of the Tolman school. But I shall make more of this point in a

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19. It may be suggested that the dilemma could be avoided by redefining "x has a conditioned eyelid response habit" so that the antecedent would also contain "x is deprived of food". But this would not remove the dilemma. It would still be true that the disposition defined by 3.2, whatever we might choose to name it, would be a disposition which would be acquired by classical conditioning.

later section of this introduction. Since I am primarily concerned with S-R psychology and spatial learning I want now (1) to revise the scope of these assumptions so that they will be restricted to simple T-maze learning and (2) to examine the evidence for these revised assumptions.

7. S-R Psychology and Spatial Learning.

It should have become clear by now that the two assumptions of S-R psychology will not account for classical conditioning. The question I would like now to consider is whether they will account for simple T-maze learning. I will argue in this section that they do not and that we must abandon these assumptions if we are to understand simple spatial learning. The arguments which I will present against them and the alternative theoretical program which I will suggest, constitute, in my mind at least, the primary tenets of the Tolman school.

Let us begin with a description of a simple T-maze experiment. A rat is deprived of food, say, for 24 hours, and then is taken from the home cage and placed on the starting path of a single unit T-maze. On, say, the west end, F, of the T there is a small cup containing wet food, while on the east end, B, there is a similar cup which is empty. The rat is allowed to stay in the maze until it reaches the food at F and eats, say, for 20 seconds. After

this the rat is removed from the maze and returned to the home cage, where later it is fed the rest of its daily ration. On the following night when the rat has again been deprived of food for 24 hours, it is again placed on the starting path of the maze and the same procedure is followed. At the end of several nights of such training we observe that when the rat is placed on the start of the maze at t it runs immediately to F, avoiding the blind alley, B, and arriving at F before t + e. Before this training, that is, on the first trial, when the rat was first put on the start of the maze it spent a long time exploring the starting path, the choice point, and the two paths leading to F and to B. On subsequent trials, however, the rat spent less and less time in reaching F, and entered the path to B less and less frequently, until finally it began to run directly to F with great speed on every trial. Now how are we to describe this change in behavior?

The reader by now can probably anticipate my answer to this question. Such training produces a disposition in the animal to run directly to F when deprived and put on the maze. Let me write the conditioned definition for this disposition before I discuss it.

6.0 x is deprived of food for N hours before t, and x is on start of maze at t and x eats no food between t and t + e $\supset$ (x runs directly to F, avoids B, and arrives at F before t + e $\equiv$ x has a disposition to run directly to F, when deprived and put on the maze).

Now there are several things about this definition that require comment. (1) The matrix "x is deprived etc." must be included in the antecedent because, as in the case of the conditioned salivary response habit, we find that when the animal is satiated it does not make the required response. Under these conditions we cannot decide whether the animal has the disposition or not. Because of this we include "x is deprived etc." in the antecedent. Hill in his Principles of Behavior (8) vaguely recognizes this point, namely, that the habits or dispositions with which he is concerned are related in some way to motivation. Thus he says :

> It is quite impossible to predict the vigor or persistence of a given type of action from a knowledge of either habit strength or drive strength alone. (8)

Definition 6.0 solves this problem in the same way that definition 5.1 solved it for the conditioned salivary response habit. (2) Exactly the same reasoning applies to the inclusion of the matrix "x eats no food etc." in the antecedent of 6.0. (3) The specification that the animal reach F before t + e is included in the definition for the same reason that it was included in all the previous definitions--to distinguish between "chance" and habitual responses. Consider for a moment an animal which has never been trained on this maze and is now put on the starting

path for the first time. Suppose further that he goes directly (as a result of a position habit) to F. If the animal has not been trained on similar mazes it is very unlikely that he will reach F before t + e, although he does go to F directly. Thus this temporal specification will enable us to deny that the animal has the disposition. On the other hand, if he has been trained on other similar mazes, he may reach F before t + e, and in this case we will assert that he has the disposition. How he acquired the disposition can then be answered by an analysis of the similarity of this maze to the ones on which he was trained.

With these comments made, we may, I hope, be able to agree that definition 6.0 conforms with the notion of "response sequence", as well as any relatively precise notion can conform with a vague one. In short, I believe that what the S-R psychologists have meant by saying that an animal acquires a response sequence is that it acquires a disposition to run down the practised path, when deprived and put on the path. Since the acquisition of this disposition constitutes for them the learning of the maze, we may write the following explicit definition :

6.1 x has learned the maze ≡ x has a disposition to run directly to F, when deprived and put on the maze.

From what I have said so far Hull's principle of primary reinforcement seems much more appropriate to T-maze

learning than it does to the conditioned eyelid response habit. Let us, then, try to write a first law of maze learning which conforms with this principle and see if it is true. An analysis of Hull's principle (see page 32) indicates that there are two important conditions which must be satisfied if an animal is to learn a maze. First there must be contiguity between the stimulation received at the choice point and the response of going to F. And second there must be contiguity between the response of going to F and, as Hull says, "diminution in the receptor discharge characteristic of a need". In other words, any training procedure in which the following conditions are satisfied should, according to this principle, produce learning. (1) The animal must be deprived of food before being given the training. (2) The animal must run over the choice point in order to get to F. (3) There must be food at F whenever the animal goes there. And (4) the animal must eat the food when he gets to F. Thus, I propose the following law of maze learning as the one which accords with Hull's principle :

13.0 For all x and all t, x is given N trials in each of which x is deprived of food N hours before t, x is put on start of maze at t, x goes over choice point, and x eats at F for N seconds $\equiv$ x has learned the maze.

Now is law 13.0 true? There certainly seems to be no evidence against the assertion that if an animal is given

Hull's training then it will learn the maze. In fact there is a great deal of evidence drawn from many kinds of experiments which supports this implication of law 13.0. This abundance of evidence is very impressive and is perhaps the reason why many American psychologists have accepted Hull's principle and have given little consideration to the criticisms which have been directed against this principle by the Tolman school.

Tolman's criticisms have not, however, been directed against this first implication of law 13.0, but against the second implication which asserts that if an animal has learned the maze then it must have been given Hull's training. The latent learning experiments, Tolman clains, provide unambiguous evidence against this latter assertion. Now there are three different kinds of latent learning experiments, and although they differ from each other in procedure, they all demonstrate that rats will acquire the disposition to run directly to F when deprived, even though they have not been given Hull's training.

(1) The first kind of latent learning experiment is historically the earliest.[20] It was first performed by Blodgett (1) and later by Tolman and Honzik (21) and others.

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20. It should be noted that all of these early latent learning experiments were performed in multiple rather than single unit mazes. This difference, however, is probably not important for our present discussion.

In such an experiment rats are given N trials in each of which they are deprived of food, put in the maze, and allowed to wander through it until they come to F. Since there is no food at F they cannot eat at F during these trials. Consequently as soon as they reach F, they are removed from the maze. After N unrewarded trials, the rats are given one trial of the Hull procedure in which they go through the maze, find food at F and eat. On the next day, when deprived and put on the start, they run directly to F. This is evidence against law 13.0 because these rats acquired the disposition to run directly to F when deprived, although they have only been given one trial of Hull's training.

Since the significance of the latent learning experiments has frequently been misunderstood it is important to stress their relevance to law 13.0. B. Reynolds (12), for example, has recently performed a repetition of Blodgett's experiment in which he concludes that no latent learning took place. His evidence for this conclusion is that most of the learning occurred in the unrewarded trials, and consequently there was no "latent learning as manifested by a sudden drop in the error curve upon the introduction of reward". (12) Now the differences between Blodgett's and Reynold's results are, of course, important, but these differences are not, I think, relevant to our present discussion.

Thus, despite the differences between them, the results of both experiments provide convincing evidence against law 13.0.

(2) The second kind of latent learning experiment was performed by Buxton (4). His procedure was very similar to Blodgett's except that (A) the rats live in the maze for 18 hours a day instead of being run through it one trial a day, and (B) just before the test trial the rats are carried by hand and fed at F, instead of being allowed to run through the maze and find food at F. Then when they are put at the start of the maze they run directly to F although they have received no trials of Hull's training.

(3) The third kind of latent learning experiment was performed by Spence and Lippitt (18). The rats were given N trials in each of which they were satiated with both water and food, and there was water at B and food at F. They were forced down path to B and the path to F an equal number of times, and each time they reached either B or F they were removed from the maze. All rats that either drank at B or ate at F were discarded from the experiment. On the test trial one group of rats was deprived of food, the other deprived of water, and then placed at the start of the maze. A large percentage of the rats in the first group ran directly to F, and a large percentage of those in the second group ran directly to B.

It should be clear now that law 13.0 and Hull's principle of primary reinforcement must be abandoned. The theoretical problem that faces us now is exactly like the one which we discussed when we considered law 11.1 (see pages 20-21) for the conditioned eyelid response habit. Thus the discovery of latent learning is the discovery of several (at least three and perhaps more) procedures which will produce the disposition to run directly to F when deprived and put on the start of the maze. We would like to find some characteristic which all these procedures, along with Hull's, have in common, so that we could then write a law which would have the same form as 13.0, but which would be true. So far little has been done in this direction. The reason for this lies in the fact that very little is known about the factors which are necessary for latent learning. Only after we have experimentally investigated these in some detail and have explored other possible procedures, will it be fruitful to attempt to state a general law of maze learning. In conclusion, then, it should be emphasized that the Tolman school does not claim to have stated such a law, it only claims to have provided evidence that law 13.0 is false.

I pointed out in the preceding section that S-R psychology was based upon two empirical assumptions : (1) that the principle of primary reinforcement (ie., law 13.0) was

true, and (2) that all the dispositions acquired in learning situations are essentially similar. Let us now consider this second assumption when it is restricted to simple T-maze learning.

Now that we have given up the vague notion of "S-R connection" we shall have to state this assumption in another way. Let us return to the disposition defined by definition 6.0, the disposition to run directly to F when deprived and put on the maze. Such dispositions are readily acquired whenever animals are given the appropriate training. Everyone must surely agree to this. But this is not what is asserted by the second assumption of S-R psychology. What this assumption asserts is that this disposition is the only disposition which is acquired in a T-maze situation. Although such an assertion obviously requires evidence for its support, few S-R psychologists have recognized this. The Tolman school, on the other hand, has consistently attacked this dogma and has emphasized that there may be many different kinds of dispositions acquired in even such a simple situation as the T-maze.[21]

Much has been written by others about Professor Tolman's "expectancy theory", but no one has stressed what I believe to be its main virtue, namely, its insistence that

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21. In a paper called "The acquisition of string-pulling by rats--conditioned response or sign-gestalt?" (20) Tolman has named and discussed seven different kinds of learning.

many dispositions may be acquired as the result of a particular kind of training in a particular situation. This idea is, I believe, implicit in his contrast between "generalized means-end-readinesses" and "particular means-end-readinesses".[22] Each of these particular means-end-readinesses correspond to what I have called "particular dispositions". Thus, for example, an animal trained in a particular maze with food at F may acquire (1) the disposition to run down the practiced path to F, (2) the disposition to short-cut to F, and (3) the disposition to run to the side of the room where F is located. Each one of these dispositions, although different from every other one, belongs to a more generalized means-end-readiness in the sense that if the animal has one it presumably has all the others, and if the location of the food is changed, each of these dispositions will change appropriately. It is this generalized means-end-readiness which Tolman regards as the "sign-gestalt"[23]

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22. These notions occur in many of his papers. They are treated in some detail in his book Purposive Behavior in Animals and Men (19) and an article with Krechevsky called "Means-end-readiness and hypothesis--a contribution to comparative psychology". (22)

23. Tolman's view is not always consistent and in some passages he seems to think of the particular means-end-readiness as a "sign-gestalt". The following quotation, however, seems consistent with my interpretation. "What we are calling sign-gestalt are in fact no more than direction-distance correlations with, however, a further emphasis upon their whole-like or 'gestalted' character." (19)

or "expectation".

If my interpretation of Tolman's expectancy theory is correct, then the main point of this theory lies in the insistence that before we attempt to state any laws of learning we must first discover what other dispositions are acquired by a particular kind of training and then discover how these dispositions are interrelated. The purpose of the experiments in this present paper is simply (1) to discover if training in a single unit T-maze produces also a disposition to short-cut to the goal location and (2) to discover some of the characteristics of this disposition.

8. The Short-cut Disposition

In a very important chapter[24] in Purposive Behavior in Animals and Men (19) Tolman describes a number of different experiments which illustrate "the rat's propensity after learning, to select certain types of spatially directed or arranged paths more readily than others". (19) The first of these experiments is one reported by Dashiell in which he used his "checker-board" maze. In this maze there are 20 equally long paths from the start to the goal. "Each animal," according to Dashiell, "learned soon to adjust to the situation not by fixating some particular pathway but by running in the general direction of the exit now by one succession of turns and now by another." (19) Next Tolman

cites some of the many T-maze experiments in which it has been discovered that goal-pointing blinds are entered more frequently than blinds which point in the opposite direction. Tolman then describes two experiments by Yoshioka. The first of these presented the rats with two equally long alternative routes to the goal, but path A turned back towards the goal sooner than path B. After training Yoshioka discovered that the rats tended to choose path A rather than path B. In the second experiment, Yoshioka demonstrated that rats tend to hug that wall of an open space which is nearer the location of the goal. Finally in some footnotes Tolman refers (1) to Lashley's (9) discovery of two rats that climbed out of the alley of the maze and took the direct route back to the food box, and (2) to a similar observation of Helson's (6).

All these experimental results suggest that training in a maze problem produces a variety of dispositions. It is not clear however from these results just what dispositions are acquired. A number of various maze studies have attempted to isolate these factors by statistical techniques. However, little has been done, until recently, to isolate these dispositions experimentally. In a recent paper by Tolman, Ritchie and Kalish (23) the short-cut disposition was isolated experimentally.[25]

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25. In discussing the experiments drawn from the Tolman, Ritchie and Kalish series called "Studies in spatial learning", I will not necessarily interpret their results in precisely the same way that they do in those papers.

Fifty-six female rats were trained for 4 days to run over a circular table-top 3 feet in diameter, through an alley, and down an L-shaped path to food boxes containing food. Behind these food boxes was a desk lamp containing a 5-watt bulb. This was the only illumination in the room. On the fifth day the L-shaped path and the food boxes were removed. A block was placed at the end of the alley and 12 six-foot paths were placed around the table-top. These paths began at a point 90 degrees to the right of the alley and radiated in a counter-clockwise fashion, each path being placed 10 degrees to the left of its neighbor. One test trial was then given each of the 56 rats. On this trial the rats were placed at the start, allowed to run into the blocked tunnel, come out, and wander about the table-top. The rats were removed after they had chosen one of the 12 paths and had run to the end of it.

On the test trial three of the 56 rats were discarded when they failed to make a choice after six minutes. Of the remaining 53 rats 36 percent chose the path which ended at a point four inches to the left of the place where the food box entrance had been during the pre-test trials. The remaining rats were distributed in a "random" fashion over the other 11 paths.[26]

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26. One exception to this statement should be made. The path which was 90 degrees to the right of the alley, and thus was both the last path on that side, and also the path which was parallel to the original path, was chosen by 17 percent of the 53 animals. The significance of this result will be discussed in a later section where similar results were discovered in a later short-cut experiment.

After presenting these results the authors try to answer the criticism that the rats which took the short-cut "merely ran towards the light, a response which was rewarded during the pre-testing training." (23) The gist of their answer (although it is not expressed in this way) was that the disposition to take the most lighted path when the original path is blocked is not the same disposition as the disposition to take the original lighted path. Although I think this argument is a good one, I doubt whether it will satisfy those who have raised the criticism. The following fact, discovered in a later short-cut experiment, should, however, make this argument more understandable. Here several different training techniques were explored in an attempt to discover whether rats would short-cut from the opposite side of the choice point.[27] Three training techniques had to be discarded before one was discovered which gave unambiguous results. Before discovery of this latter technique, a total of 56 rats were run on a simple T with lights at both ends, one at F and the other at B. Out of the 56 animals that had been trained in this situation only one rat took a path which pointed towards a light on the test trial. Now if the learning of the original T-maze consisted merely in learning to run down one lighted path and to avoid the other one, then on the test trials the rats should have

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27. These techniques will be described in detail in a later section of the paper.

chosen the lighted paths more frequently than the non-lighted ones.[28] The fact that all but one of the rats avoided the lighted paths indicates that the disposition to run down a practiced path to a light is not the same as the disposition to run down a lighted path when the original path is blocked. Only when certain other conditions are fulfilled will the rats run down the lighted short-cut paths. One of these conditions it has been discovered is that the rat run first into the blocked alley. Thus the presence of the lighted paths is not by itself sufficient.

Let us turn now to the problem of writing a definition for this short-cut disposition. The following definition is the one proposed by the authors in the paper which we have been discussing.

7.0 x is deprived of food and x has been trained on path P and x is now put on path P and P is now blocked and there are other paths which lead away from path P, one of which points directly to location L $\supset$ (x runs down the path which points directly to location L $\equiv$ x expects food at location L).

I no longer believe that this definition is a good one. Let me list what I think are its faults. (1) To include the matrix "x has been trained on path P" in the antecedent is unnecessary, and may violate the empirical requirement

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28. Of course one would not expect on this hypothesis that the rats would necessarily chose the correct lighted path, but merely that they would avoid all paths which were not lighted.

discussed on pages 21-26. (2) The antecedent should include "x is put on path P at t, and x runs into the alley before t + e". (3) It is misleading to make "x expects food at location L" equivalent to "x has the disposition to short-cut to L, from P, when P is blocked etc." Each of these remarks needs a fuller explanation.

Let us begin with the matrix "x has been trained on path P". This matrix was included in the original definition because it was believed that it would afford a way of distinguishing "chance" from habitual short-cuts. There are two reasons why this belief was mistaken. In the first place, this matrix was vacuous since the kind of training is left unspecified. In order to make the matrix effective in distinguishing "chance" from habitual short-cuts, one must specify in the antecedent the kind of training which, in fact, will teach the animal to short-cut when path P is blocked. But such a definition quite obviously violates the empirical requirement that there be no empirical relation between the first and second matrices of a conditioned definition.[29] Thus with such a definition we would be unable to distinguish rats that had the disposition from those that

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29. Such a definition would enable us to distinguish animals that had the "short-cut ability" (ie. the disposition to short-cut when given this training) from those that do not have this ability, but this is not the disposition with which we are concerned.

did not. As a consequence of these considerations we must search for a more effective and less objectionable method of distinguishing "chance" from habitual short-cuts.

By specifying the speed with which the rats run into the blocked alley, we may be able to achieve this end. This is the reason why I am suggesting that we include in the antecedent, the matrix "x is put on path P at t and x runs into the alley before t + e". There are at least two reasons why this way of distinguishing "chance" from habitual short-cuts may be a good one. (1) We have discovered, in a later short-cut experiment, that only those rats which run directly into the alley, will choose the short-cut. Those that show hesitation and explore other paths before going into the alley rarely if ever take the short-cut.[30] (2) Common sense would assert that a rat which chose the short-cut path was trying to get to L.

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30. Although this may appear to be, this is not a violation of the empirical requirement. Let "p" mean "x runs into the alley" and "q" mean "x takes the short-cut". Now our definition is

$$p \supset (q \equiv r)$$

while the empirical relation we have discovered is

$$-p \supset -q.$$

Thus we have discovered that when "-p" is true "-q" is true, and that when "-p" is false (ie, when "p" is true) "-q" may be either true or false. But according to our definition "$q \equiv r$" is true only when "p" is true (ie, when "-p" is false). Therefore, when "$q \equiv r$" is true, "-p" is false, and "q" may be either true or false. Thus the empirical requirement is satisfied.

On the other hand if these conditions were not satisfied common sense would not be able to decide whether the rat had this disposition or not.[31] Thus my proposed definition for the matrix "x has a disposition to short-cut to L" conforms in this requirement with common sense usage.

My final criticism of definition 7.0 concerns the words we should use in "R(x)" the matrix which is being introduced. On page 49 in discussing Tolman's expectancy theory I said that Tolman tends to restrict the notion of "expectation" to the more generalized means-end-readiness or disposition. If we are to conform to this usage it would be best not to call this particular short-cut disposition "an expectation of food at location L". The reason why the authors of definition 7.0 called the short-cut disposition "an expectation" was that they were concerned with showing that this particular disposition is more analogous to the disposition a human has when common sense says "that man expects food at L" than is the disposition to take the practiced path. This is an important point, but to make this point it is not necessary to call this disposition "an expectation". I now think it will be

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31. I am not defending this loose way of talking, or arguing that this behavior is evidence that the rat knows the alley is blocked etc. All that I am trying to show is that the way common sense uses the expression "x has a disposition to short-cut to L" is close to my proposed usage of the expression.

better to wait until we have discovered some of the laws relating the various dispositions which are acquired in a simple T-maze. We may then find it useful to introduce the expression "x has an expectation of food at location L" as an abbreviation for a longer expression like "x has disposition D_1 and D_2 and D_3 or x has disposition D_4."[32]

This concludes my criticisms of definition 7.0. The following, then, is my proposed definition for the short-cut disposition, discovered in the experiment, described in the beginning of this section (see pages 51-53)

7.1 x is deprived of food for N hours before t and x is on path at t and x runs directly into the alley before t + e and path P is blocked and there are other paths which lead away from path P, one of which points directly to location L $\supset$ (x runs down the path which points directly to location L $\equiv$ x has a disposition to short-cut to L when deprived, when the practiced path is blocked and when x runs into the block).

I want now to turn to a description of the second experiment in the Tolman, Ritchie and Kalish series (24), because this experiment set the problem for my experiments in spatial learning.

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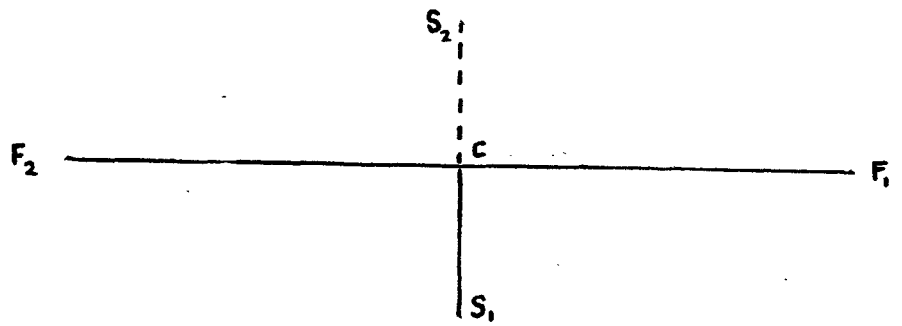
32. This is presupposing that we have discovered a law like 14.0

14.0 For all x, x has training $T_1 \equiv$ (x has D_1 and x has D_2 and D_3) v x has D_4.

9. Two Places Versus One Place

The second experiment of the spatial learning series was designed to determine which of the following two problems is the easier to learn (1) to go to one place by two different routes, or (2) to go to two places by two different routes.

The apparatus used was an elevated maze (see schematic drawing below) made of two inch painted pine paths. The



center path running from the food boxes at F_1 to those at F was eight feet long, while the paths S_1C and S_2C were each two feet long. This maze was set in a room which was 30 by 45 feet. The F_1-F_2 path ran north and south, F_1 at the north, and F_2 at the south, 15 feet to the north of the home cages. The only illumination in this room came from three overhead lamps placed symmetrically about the choice point, and parallel to the F_1-F_2 path approximately 8 feet

to the east of the F_1-F_2 path.

Two groups of eight rats were run on this maze. The experimental group were required to turn right when facing, say, west (ie. when started at S_1) and to turn left when facing east (ie. when started at S_2). The control group, on the other hand, were required to turn right on all trials. Each group were given three trials from S_1 and three from S_2, every evening. The correction method was used with both groups.[33]

All of the rats in the experimental group reached the criterion (ten successive errorless trials) within eight trials or less. The rats in the control group, on the other hand, were run for seventy-two trials (or twelve nights) and only three of the eight rats reached the criterion[34] within that time, while the rest developed habits of always going to the same place (either F_1 or F_2).

From these results we may conclude that the problem

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33. When an error was made the rat found the door to the food box locked and was required to turn around and run to the opposite food box. The order of starting positions was $S_1S_2S_2S_1S_1S_2$ and $S_2S_1S_1S_2S_2S_1$ on alternate days. The rats in the experimental group were given a position habit test before the training series, and then the problem set against this habit.

34. Two of these rats reached the criterion in fifteen trials, and the other in twenty-two trials.

set for the experimental group is easier than the one set for the control group. Let us now consider the following question. Why is it easier for rats to learn to go always, say, to the north, than it is for them to go north when run from the east and south when run from the west?

Although I do not intend to answer this question now, I do want to indicate what I believe is the correct way to answer it. Now the explanation which I am about to propose is based upon several untested empirical assumptions.[35] Should these assumptions turn out to be true, then they would comprise the explanation for which we are searching. This point is an important one since the custom among contemporary behavior theorists of the S-R school is to formulate their "explanations" in terms of untestable assumptions, or assumptions which they say cannot be tested directly.

But before I can state these assumptions I must introduce a new disposition predicate. Suppose that we trained rats on the short-cut apparatus described on pages 52-54 with the circular table-top and 18 radiating paths. And suppose that during this training there is food on the path which points directly to F_1 and no food on the other 17 paths. After several trials of this training the rats will learn to run <u>directly</u> from the start to the path which points directly to F_1, and out on this path to food. Let

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35. The purpose of the experimental part of this paper is to test some of these assumptions.

us call the disposition acquired in this situation "the disposition to take the shortest path from S_1 to F_1" and use the following conditioned definition to introduce this expression :

8.0 x is deprived of food for N hours before t and x is on path at t and there are paths which lead away from path P, one of which points directly to $F_1 \supset$ (x runs down the path which points directly to $F_1 \equiv$ x has a disposition to take the shortest path to F_1 when deprived).

Now the only difference between this disposition and the short-cut disposition, defined by 7.1, is that definition 8.0 does not include in the antecedent the condition that the rat run into the blocked alley before t + e and before choosing the path to F_1. Thus, the short-cut disposition is a special kind of shortest-path disposition, and it follows logically that all rats that have the short-cut to F_1 disposition have the shortest-path to F_1 disposition.[36]

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36. Let "P_1" mean "x is deprived of food for N hours before t and x is on the path at t and there are paths which lead away from path P, one of which points directly to F_1", and let "P_2" mean "x runs into alley before t + e and path P is blocked." Then "$P_1 . P_2 \supset (Q \equiv R_1)$" is the definition of the short-cut disposition and "$P_1 \supset (Q \equiv R_2)$" of the shortest-path disposition.
Now we see that "R_1" and "R_2" are both equivalent of "Q" whenever "$P_1 . P_2$" is true (since "$P_1 . P_2 \supset P_1$" is analytic). Thus whenever we can determine the truth value of "R_1", it will have the same truth value as "R_2", since under these conditions both "R_1" and "R_2" are equivalent to "Q". But whenever only "P_1" is true

We may now turn to the proposed explanation of the results of the two place versus one place experiment. This explanation consists of a set of assumptions or postulates which entail that the training given the experimental group will produce rapid learning, and that the training given the control group will produce conflicting dispositions and thus make learning difficult. I will now list these assumptions, and in order to make clear the logical relations between them and the theorems derived from them, I will write them also in logical notation.

15.0 $(x)\ P(x) \supset Q(x)$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to short-cut from S_1 to F_1.[37]

15.01 $(x)\ Q(x) \supset R(x)$

For all x, x has a disposition to short-cut from S_1 to $F_1 \supset$ x has a disposition to take the shortest path from S_1 to F_1.[38]

(36. continued) we cannot determine the truth value of "R_1", although we know the truth value of "R_2". In short then, we say that "$R_1(x) \supset R_2(x)$" is analytic, because if we can determine the truth value of "$R_1(x)$" the truth value of "$R_2(x)$" must be the same.

37. The evidence from the short-cut experiment which I have discussed on pages 52-54 suggests that 15.0 may be true.

38. This proposition is analytic, as explained in footnote 36.

15.1 $(x)\ R(x) \supset S(x)$

For all x, x has a disposition to take the shortest path from S_1 to $F_1 \supset$ x has a disposition to turn right at C from S_1.[39]

15.2 $(x)\ R(x) \supset T(x)$

For all x, x has a disposition to take the shortest path from S_1 to $F_1 \supset$ x has a disposition to take the shortest path from S_2 to F_1.[40]

15.3 $(x)\ T(x) \supset U(x)$

For all x, x has a disposition to take the shortest path from S_2 to $F_1 \supset$ x has a disposition to turn left at C from S_2.[41]

Now if all of these assumptions are true we can explain the results of the two place versus one place experiment, since we can show that training, say, from S_1 to F_1 reinforces the disposition to turn left at C from S_2, which is a correct response for the experimental group, but an error for the control or two place group. The demonstration is not difficult. The reader can prove for himself that these

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39. At present there is no evidence for 15.1. One way of testing this assumption would be to train rats with the circular table top and radiating paths, and then give them a critical trial with the radiating paths removed, and a T-maze choice at C.

40. There is no evidence for 15.2. This could be tested by training rats with the radiating paths from S_1 and then testing them on radiating paths from S_2.

41. There is no evidence for 15.3. It could be tested in the same way as 15.1.

assumptions entail the following two laws :

16.0 $(x)P(x) \supset S(x).U(x)$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to turn right at C from S_1 and left at C from S_2.

and

16.1 $(x)\acute{P}(x) \supset S(x).U(x)$

where "P(x)" means "x is trained from S_2 over choice point C to F_1." [42]

Now since the experimental group is trained from both S_1 and S_2 to F_1 we can combine 16.0 and 16.1 to get :

16.2 $(x)P(x).\acute{P}(x) \supset S(x).U(x)$ [43]

Let us now turn to the two place or control group which

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42. The following is the form of the proof for 16.1 where "W(x)" means "x has a disposition to short-cut from S_2 to F_1." :

(1) $\acute{P}(x) \supset W(x)$
(2) $W(x) \supset T(x)$
(3) $T(x) \supset U(x)$
(4) $T(x) \supset R(x)$
(5) $R(x) \supset S(x)$

Therefore : $P(x) \supset S(x).U(x)$

The reader should note that to prove 16.1 we must assume $(x)T(x) \supset R(x)$ in step number 4, while in proving 16.0 we must assume $(x)R(x) \supset T(x)$. As a consequence we must find evidence for $(x)T(x) \equiv R(x)$, if my explanation is to be valid.

43. This follows from Leibniz's praeclarum theorema :

$p \supset q.r \supset s :\supset : p.r \supset q.s$

is trained to run to F_1 from S_1 and to F_2 from S_2. Since we know by law 16.0 that the rats trained from S_1 to F_1 acquire a disposition to turn right at C from S_1 and left at C from S_2, we have only to discover what dispositions will, according to these assumptions, be acquired as a result of training from S_2 to F_2. The reader should be able to derive the following law from these assumptions :

17.0 $$(x)L(x) \supset M(x).N(x)$$

For all x, x is trained from S_2 over choice point C to $F_2 \supset$ x has a disposition to turn right at C from S_2 and left at C from S_1.

Thus when 16.0 and 17.0 are combined to describe the training of the control group we get

17.1 $$(x)P(x).L(x) \supset M(x).N(x).S(x).U(x)$$

But "M(x)" means "x has a disposition to turn right at C from S_2" and "U(x)" means "x has a disposition to turn left at C from S_2"! There is a similar opposition between the dispositions named by "N(x)" and "S(x)". Thus we see that the control group, according to these assumptions, acquires conflicting dispositions, which accounts for the inferior performance of this group.

In conclusion then, according to these assumptions the dispositions acquired by the experimental group when trained from S_1 are the same as those acquired from S_2. As a con-

sequence the rate of learning should be very rapid since training from either side increases the dispositions to make the correct response from both sides. In the case of the control or two place group, on the other hand, the situation is quite different. According to the same assumptions, this latter group acquires opposed dispositions when trained from both S_1 and S_2. As a consequence learning in such a situation is theoretically impossible.[44] More must be known about the relative strengths of dispositions acquired in these different ways before we will have a full understanding of the behavior of the animals in this group. However it is clear that these assumptions entail that the one place group should learn much more rapidly than the two place group.

I have given what I believe is a possible explanation of the results of the two place versus one place experiment for two reasons. (1) I wished to indicate how a theory of learning which dispenses with the vague notions of S-R psychology might attempt to explain such results.[45] And (2) I wished to indicate how this experiment sets the problems

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44. The fact that 3 of the 8 rats in this group did learn the problem indicates that this is not the case. This fact might be accounted for, if we assumed that other dispositions which we have not considered may also be acquired, and that these 3 rats learned the problem on the basis of these other dispositions.

45. Perhaps it should be mentioned here that there seems to be no obvious explanation of these results in any of the current hypotheses of the S-R psychologists.

for the experiments about to be reported.

10. The Assumptions To Be Tested

The first two experiments about to be reported are designed to determine whether training from one side increases the disposition to make the correct response from the opposite side when rats are trained from two starting places to the same food place. Laws 16.0 and 16.1 derived from my assumptions state that the same dispositions are acquired from S_1 as are acquired from S_2. If these two laws are true, then rats trained from S_1 to F_1 and S_2 to F_1 should learn more rapidly than rats trained from S_1 to F_1 and S_3 to F_3 (where S is a starting position which is discriminable from S_2, and F_3 is a food place which is discriminable from F_1), since the dispositions acquired by training from S_3 to F_3 are not the same as those acquired by training from S_1 to F_1. The proof of this is simple.

16.0 For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to turn right at C from S_1 and (left at C from S_2)

and

18.0 For all x, x is trained from S_3 over choice point D to $F_3 \supset$ x has a disposition to turn left at D from S_3 and (right at D from S_4) [46]

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46. The brackets around "left at C from S_2" and "right at

Now if law 16.2 is true, each trial for the one place group from, say, S_1 increases the disposition to make the correct response from S_2 as well as from S_1. But each trial for the S_1-F_1 and S_3-F_3 group from, say, S_1 only increases the disposition to make the correct response from S_1 and not from S_3. As a consequence if these assumptions are true the one place group should learn faster than the S_1-F_1 and S_3-F_3 group. The first two experiments about to be reported test this hypothesis.

The third and last experiment to be reported is designed to test the following theorem derived from the same assumptions.

19.0 $(x)P(x) \supset T(x)$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to take the path from S_2 to F_1.

(46. continued) D from S_4" indicate that these two dispositions are vacuous in the sense that in this problem the rats are never started either from S_2 or from S_4.

PART II

THE EXPERIMENTS

EXPERIMENT I

1. Animals

Sixteen pigmented male rats, approximately 3 months old were used in this experiment. These rats were M x M's from the Tryon strains at the University of California laboratory. Four days before the beginning of this experiment, they were put on a 24 hour wet-food maintenance schedule. None of these animals had had any previous experimental training. They were fed every night at 10 PM, and all the experimental trials were run between 8 and 10 PM. When the experimental training began the 16 rats were separated into 2 groups of 8 each. The group which was required to learn two paths to the same place (ie., S_1-F_1 and S_2-F_1) was called "the experimental group". The group which was required to learn two paths to two different places (ie., S_1-F_1 and S_3-F_3) was called "the control group".

2. Apparatus

The apparatus used in this experiment (cf. Fig. 1) was an elevated maze similar to that used in the second experiment of this series (24) except that instead of having end-boxes and curtains at the ends of the center path, small paper cups, 2 inches in diameter and 1.5 inches deep, were

placed on both ends of the center path. The center path was 2 meters long, while the starting path was 40 cms.

The only illumination in the room came from an overhead frosted globe containing a 200 watt bulb (cf. Fig. 2). This lamp was suspended from the ceiling approximately 7 feet above the maze. The animals lived in home cages kept in a rack which stood against the north wall. During all the training trials the animals were taken individually from their home cage in the rack and placed on the maze. After their trial they were then returned to a cage which rested on a stand in front of the home-cage rack, (cf. Fig. 2) approximately two feet from the west wall. When all four of the rats in a given home cage had been run, they were taken from the cage on this stand and returned to the home cage in the rack to wait for the next trial.

The various positions of the maze in the room are indicated in Fig. 2. Position A was that employed during the preliminary training by both groups when position habits were being tested. Position A was also the position for the experimental group. During the experimental trials, positions B and C were the two alternative positions of the maze for the control group. The animals in this latter group were run two trials each night at position B and two at position C, while the experimental group ran all

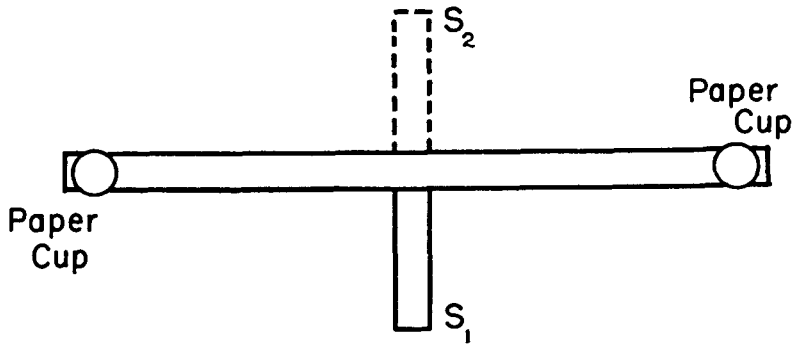


Figure 1

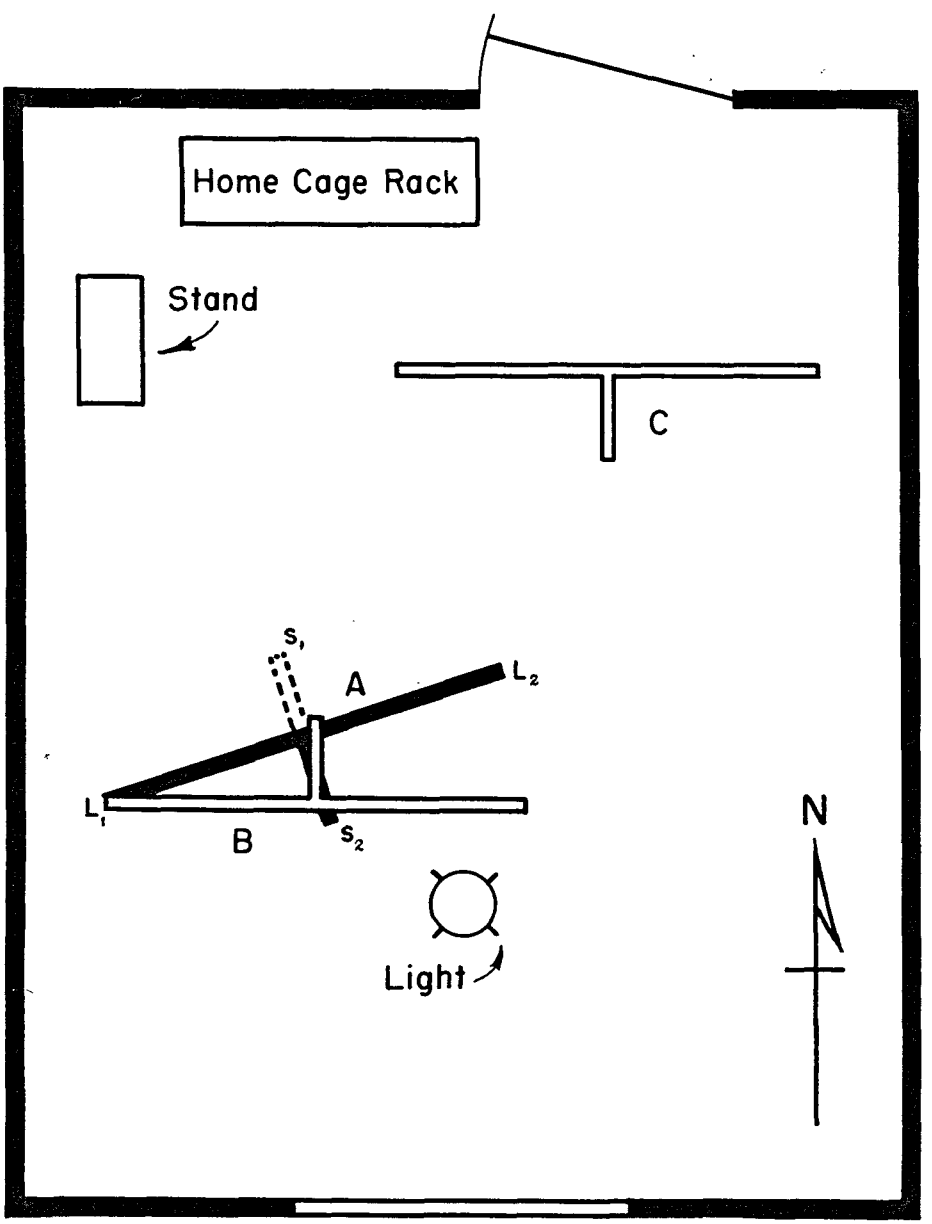


Figure 2

four of their trials in position A.[47]

During each of the trials the experimenter sat on a stool directly behind the starting path and put the rats on this starting path by hand.

3. Method

1. Preliminary Training : After 3 days on a 24 hour wet-food maintenance schedule both groups of rats were given one day's training to accustom them to run on elevated paths and to eat from paper cups on the end of such paths. This training was given during the afternoon so that the lighting would be distinct from that used later in the experimental trials. Four elevated paths were set in the room so that they were parallel to each other and ran from north to south. They were approximately three feet apart. Each of these paths was five feet long and had paper cups containing wet food at both ends. Each of the animals was given five "trials" on these paths. Such a "trial" consisted in being taken from the home-cage, placed in the middle of one of the four paths, allowed to run back and forth,

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47. For the experimental group two of these trials were begun facing south (ie., at S_1) and two were begun facing north (ie., at S_2). Thus they were required to make two right and two left turns every night. For the control group all the trials in position B were begun facing south, and all those in position C were begun facing north. Thus, the control group was also required to make 2 right turns and 2 left turns every night.

and to eat from either or both of the cups for five minutes. Thus, at the end of this part of the preliminary training each rat had had 25 minutes on some one of these four paths.

On the following night both groups were given four trials on the apparatus in position A (cf. Fig. 2) in order to determine their position habits. Two of these trials started from S_1 and the other two from S_2. The order of trials was $S_1S_2S_2S_1$. Food was placed in the cups at both ends of the center path, and the animal was allowed to remain on the path for 25 seconds after having first reached one of the two food cups. Some of the animals on some trials spent all their 25 seconds in eating from the same cup. At other times they would eat a little from one and then turn about and run to the other cup and eat there. These "retracings" were recorded along with the hesitation times which preceded each choice.

One animal in each group exhibited a consistent right-turn habit, while 2 animals in the experimental group and 3 in the control group consistently chose one side and this was later made incorrect. The other animals made no consistent choices.

2. Experimental Training : (a) Experimental Group (N=8). The animals in this group were given four trials a night for four nights on the apparatus in position A. The order of starting positions was $S_1S_2S_2S_1$ on odd-numbered

nights and $S_2S_1S_1S_2$ on even-numbered nights. The training procedure was the same as in the preliminary position habit test except that food was placed only in the cup at the west end of the path. Errors, VTE's, times, and retracings were recorded for each animal.

(b) Control Group (N=8). The animals in this group were given the same number of trials per night for four nights. Each night they were given two trials with the apparatus in position B and two trials in position C. The order was BCCB on odd-numbered nights, and CBBC on even-numbered nights. The training procedure was the same as that for the experimental group with food only in the cup at the west end of each path. Errors, VTE's, times, and retracings were recorded for each animal.

4. Results

The progress of learning in the two groups is shown in the error curve in Fig. 3. The experimental group dropped rapidly on the second day from 1.6 errors per rat on the first day to 0.2 errors on the second day, while the control group only dropped to 1.1 errors per rat on the second day. However, on the third day the control group dropped to 0.1 errors while the experimental group ran perfectly.

The mean error per rat for the four days was 2.0 errors for the experimental group with a standard deviation

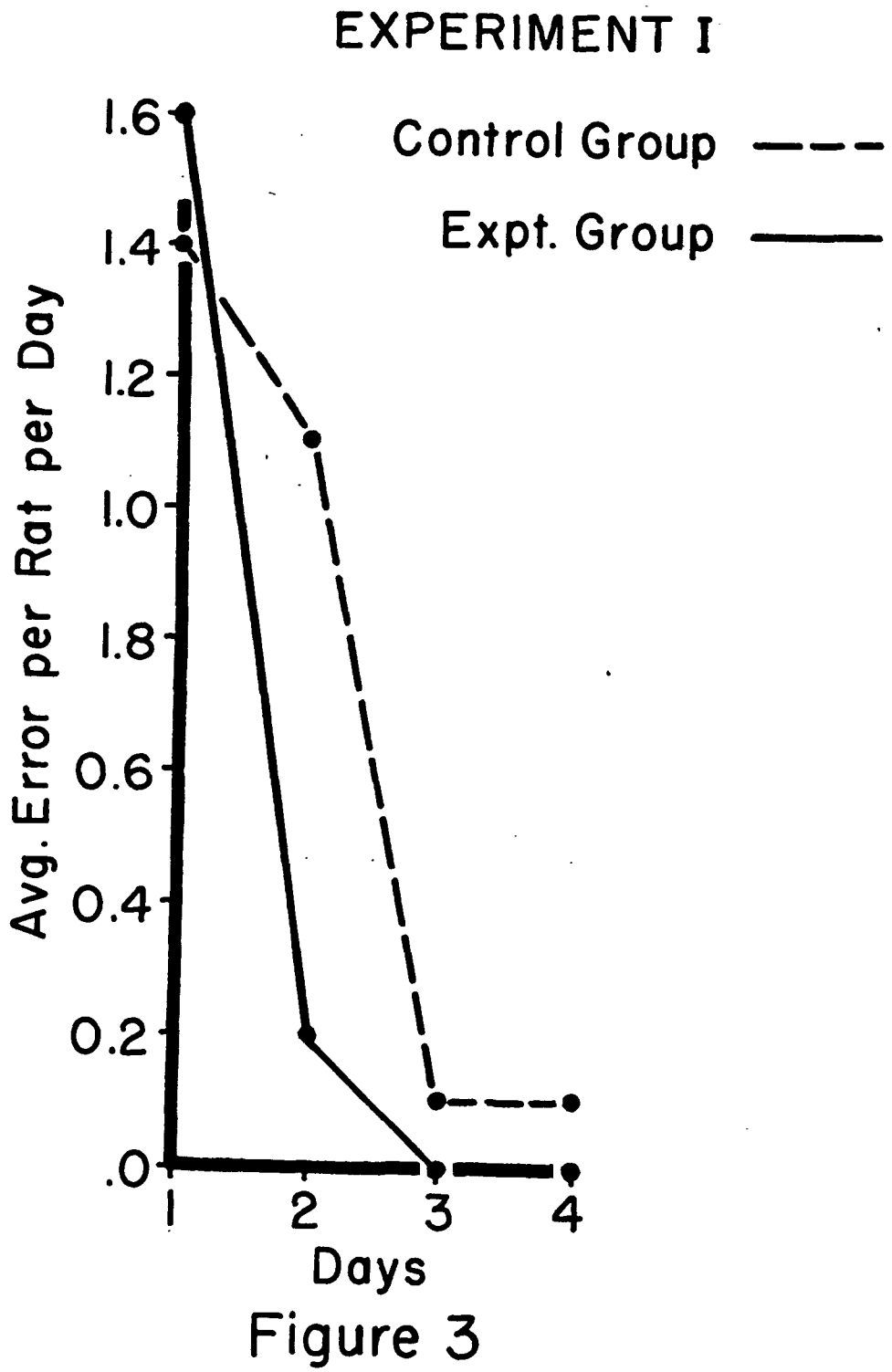

EXPERIMENT I
Control Group
Expt. Group
Avg. Error per Rat per Day
1.6
1.4
1.2
1.0
1.8
0.6
0.4
0.2
.0
1
2
3
4
Days

Figure 3

of 0.7. The mean error per rat for the four days was 2.8 errors for the control group with a standard deviation of 1.7 errors. Consequently the difference between the two groups was not statistically significant. The critical ratio of the difference between the means was only 1.15.

5. Discussion

The results of Experiment I indicate only a slight difference in favor of the experimental group, the group which learned two different paths to the same place. This difference was most marked on the second day of training. But after that, the difference between the two groups seems negligible and certainly is not statistically significant. Thus we failed to find convincing evidence for our hypothesis that if rats are trained on two starting paths which lead to the same goal place learning is more rapid than it is when the two paths lead to two different goal places.

The question that now arises is --why did we fail to get more definite evidence? There are at least three alternative answers to this question.

1) Both groups may have learned on the basis of intra-maze cues, since the center path was not rotated during the training. In this case we might expect the learning

rates of the two groups to be the same.[48]

2) The hypothesis that training from, say, S to F increases the disposition to make the correct turn from S may be false. Both groups may merely have learned to turn right when facing south and left when facing north. If this were the case, then there would be no difference in the learning rates of the two groups.

3) The final alternative answer is this : S and S , , and F and F may not have been discriminable places for the control group. If this were the case, then, according to the assumptions we are testing, training from S to F would increase the disposition to turn correctly when started from S , and thus there would be no difference between the two groups. Such a confusion of S and S and of F and F might be expected to occur if there were some strong orientational cue in the room which was the same for the mazes whether in positions B or C. The experimenter had in designing this experiment hoped to rule out this possibility, intending that the light should be the only orientational cue, and further he had hoped that the relation of the light to the goal when the maze was in posi-

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48. Although, on this hypothesis, practice from one side of the maze would increase the disposition to take the same identical path (eg., the one with the knot-hole) when started from the opposite side, this is not spatial learning in our sense of the term.

tion B would be distinct from its relation to the goal when the maze was in position C. There were, however, two possible slips in the carrying out of this intention : 1) The primary orientational cue may not have been the light but may have been some other feature of the environment which the experimenter had overlooked and which was the same for the goal locations in both of the positions of the maze (ie., B and C). Or 2) The light may have been the primary orientational cue, but the rats may not have been able to distinguish the goal locations in terms of its relation to the light.

In order to discover if any of these three hypotheses was true and thus to discover if possible why we failed to get the evidence for which we were seeking further tests with the control group were carried out. During these tests food was put at both ends of the center path so that no further learning would occur.

On the 5th night the center path was turned end for end to determine whether the performance was based upon intra-maze cues. No change in performance resulted; only one error was recorded for the entire group in four trials.

On the 6th night the experimenter tried to discover whether or not the light was an important orientational cue. To be specific, he tried to discover 1) whether the rats had learned to turn right in the lighted end of the

room and left in the dark end; or 2) whether they had learned to go towards certain specific places and away from other places, these places being specified in terms of their relation to the light. If evidence were found that the rats had acquired neither of these dispositions, then it would be clear that the light was probably not an important orientational cue. To investigate the possibility that they had acquired neither of these two dispositions the maze was moved to the positions shown in Fig. 4B. (In Fig. 4A we again see the previous positions which had been employed until day 6). Now if the rats had acquired either of these two dispositions, their behavior should break down (ie., they should stop going west) when the maze was moved to these new positions. That is, if the rats were turning right in the light and left in the dark, and they continued this, they should all go east rather than west on these test trials with the maze in the position shown in Fig. 4B. Similarly if they had learned to go to specific places and to avoid specific places in the room, they should go east when started at the south end of the room to avoid the place of the original blind, and they should also go east in the north end of the room in order to go to the place of the original goal. In conclusion then, if the rats on these test trials broke down and began going east we would have evidence that they may have acquired one of these two

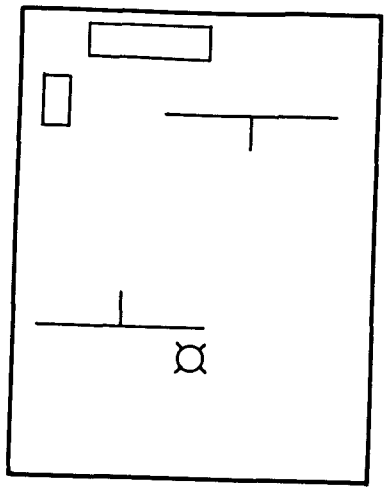


Fig. 4A
Days 1-5



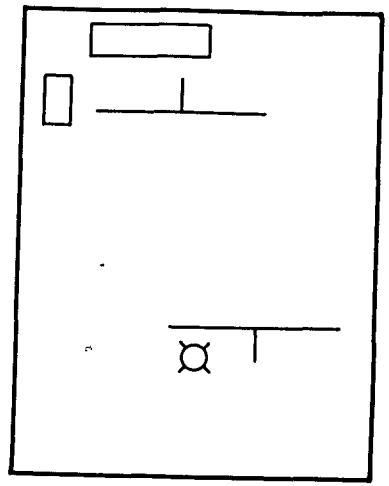


Fig. 4B
Day 6



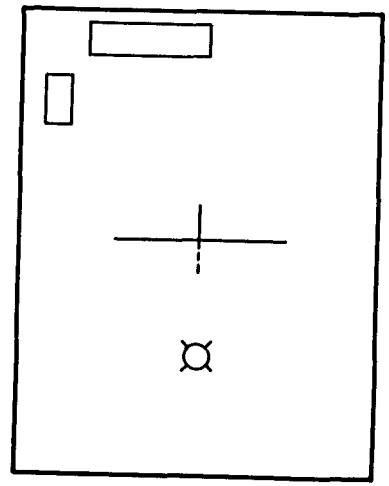


Fig. 4C
Day 7

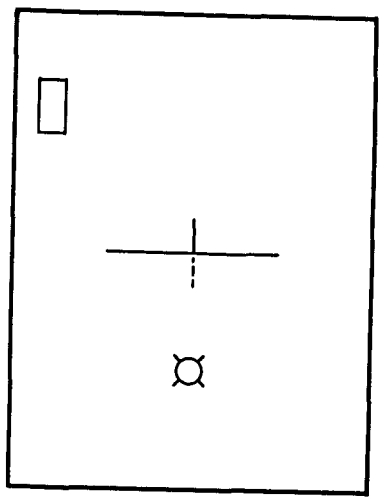


Fig. 4D
Day 8

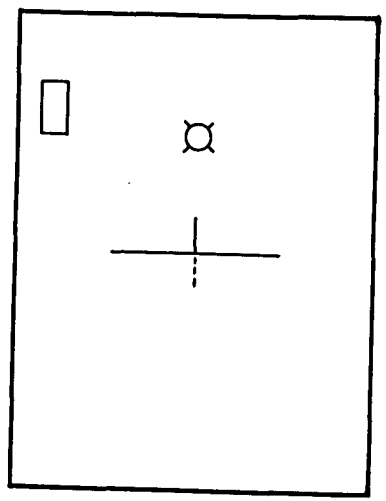


Fig. 4E
Day 9

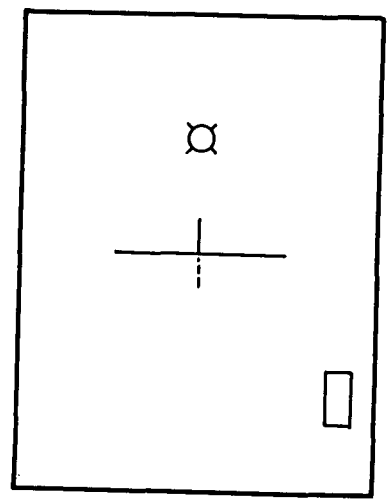


Fig. 4F
Day 10

Figure 4

dispositions. But, if they failed to break down, we would have evidence that they had acquired neither of these dispositions, and that, therefore, the light was probably not an important orientational cue. When these test trials were run there was no significant change in performance. They showed no increase in hesitation, and only three errors (runs to the east) were recorded for the entire group in their four trials. This suggests that the influence of the light was slight.

On the 7th night the rats were run with the maze in the center of the room as shown in Fig. 4C. This problem was similar to that learned by the experimental group. Only one error occurred on the 7th night. On the 8th night (Fig. 4D) the home cages of all the animals were removed to another room, and a single cage of four animals was brought into the room for each trial. After the four rats in that cage had been given a trial, they were returned to the other room, and another group of four rats was run.[49] This was done to determine the orientational influence of the home cages. No change in performance resulted on the 8th night: one error and no increase in hesitation.

On the 9th night the maze remained in the same position, but the light was shifted in position as shown in

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49. For all the following tests the home cages were kept in this other room and the same procedure followed.

Fig. 4E. This was a further and conclusive test of the influence of the light. If the light were the primary orientational cue then all the rats should go east on these trials. No change in performance resulted : 4 errors and no increase in hesitation.

At this point it became quite clear to the experimenter that there must be some strong orientational cue present in the environment which he had overlooked. He then hit upon the hypothesis that the cage which rested on the stand in front of the home cage rack (cf. Fig.2) and from which the animals were carried before each trial and to which they were returned after each trial, was probably this overlooked cue. This cage stood only 2 feet from the west wall, and since the rats in it made some noise, it might very well serve as such a cue. In short, the rats in the control group may merely have learned to run towards this cage, and since the sound coming from it was diffuse and not easily localized, the goal location (defined in terms of this extra-maze cue) was identical for the rats in the control group whether the maze was in position B or C.

To test this hypothesis this stand and cage were placed on the 10th night in the opposite corner of the room (in the south-east corner) (see Fig. 4F). Each of the rats was given 4 trials with the cage in this position. Their change in performance was astounding. There was no in-

crease in hesitation, but 24 errors out of a possible 32 were made.

In conclusion then, the reason that we failed to get the evidence for which we were searching was that the control group learned their problem as two routes leading to the cage and stand; that is, they were unable to distinguish S_3 from S_2, and F_3 from F_1. Thus the experiment failed because we failed to set the correct kind of problem for the control group. Consequently, we decided that if we could remove all the orientational cues but the light, and if we could make the two goal locations for the control group distinct, we should be able to get the evidence for which we were looking. With this end in mind, Experiment II was designed.

EXPERIMENT II

1. Animals

Fourteen pigmented male rats of the same age and stock as those used in Experiment I were used in this second experiment. They were put on the same maintenance schedule and were fed every night at 10 PM. None of them had had any previous experimental training. The one-place group (N=6) is called "the experimental group". The two-place group (N=8) is called "the control group".

2. Apparatus

The mazes used in the present experiment were exactly like the ones used in Experiment I except that the center paths were 2.5 meters rather than 2 meters and the starting paths were 50 rather than 40 centimeters.

Two rooms and two mazes were used for the control group, while only one room and one maze was used for the experimental group. The mazes were set in the rooms so that the choice points were in the exact center of the rooms. The only illumination in Room A (where the experimental group had all its trials) came from the same overhead lamp as was used in Experiment I (cf. Fig 5A). The control group were given two trials a night in Room A, starting always from S_1, and two other trials in Room B from $\acute{S}_2$, (cf. 5B) where there were two overhead lamps along the axis of the starting path.

The animals lived in cages in still a third room and were carried during the experimental training trials from this third room to one of the experimental rooms, given one trial, and returned to the home cage to wait for their next trial. The cage in which they were carried was always set exactly behind the starting path, in order to remove cage cues which had compromised the results in Experiment I.

During each of these trials the experimenter sat on a stool directly behind the starting path and put the rats

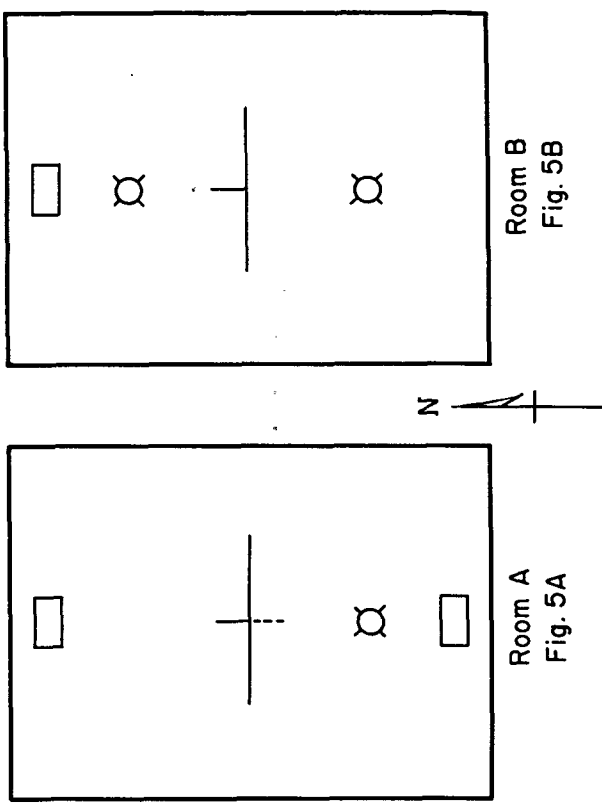


Figure 5

on the path by hand. In addition the center path was turned end for end at the conclusion of the 2nd trial in each nightly run of four trials. This was done to remove the influence of intra-maze cues.

3. Method

1) Preliminary training : After three days on a 24 hour wet-food maintenance schedule both groups were given three days training to accustom them to run on elevated paths and to eat from paper cups at the ends of such paths. This training was given in the third or home-cage room where the lighting was distinct from that of either of the two experimental rooms. They were run four times a night on five foot elevated paths which had food at only one end. During each run they were allowed to remain on the path for 60 seconds after they had run to an end and had begun to eat from one food cup. Thus at the end of the three days of preliminary training each of the rats had had approximately 12 to 15 minutes on elevated paths.

On the following night the animals were divided into two groups, the experimental, or the one-place group, and the control, or two-place group. The animals in both groups were given 4 trials in order to determine their position habits. The experimental group were tested on the maze in

Room A, with food at both ends of the center path. Two of these trials started from S_1 and two from S_2. The order of trials was $S_2S_1S_1S_2$. The animals were allowed to stay on the path for 25 seconds after having reached one of the two food cups. The control group was tested two trials on the maze in Room A from the starting position $\acute{S}_1$ and two trials on the maze in Room B from the starting position $\acute{S}_2$. The order of the trials was $\acute{S}_2S_1S_1\acute{S}_2$, with food at both ends of both center paths.

The results of this position habit test were as follows : Two of the 14 animals showed a consistent turn habit, both making 4 successive left turns, and both of these animals were in the experimental group. Two of the remaining animals showed the consistent direction habit (one going west, the other going east on all 4 trials), and both of these animals were in the control group. The remaining ten rats made unsystematic choices.

2) Experimental training : (a) Experimental group (N=6). The animals in this group were given 4 trials a night for 10 nights on the maze in Room A. The order of starting positions, and the training procedure was the same as that for the experimental group in Experiment I.

(b) Control group (N=8). The animals in this group were also given 4 trials a night for 10 nights. Two of these trials were from S_1 in Room A and two from $\acute{S}_2$ in Room

B. Thus the training procedure was the same as it had been in the preliminary position habit tests except that food was placed in only one of the two cups on the center path.[50]

4. Results

The progress of learning in the two groups is shown by the error curves in Fig. 6. Both groups dropped rapidly until on the third day when the control group reached a plateau of 0.5 error per rat. The experimental group on the other hand continued to improve until they ran four successive errorless days.

The mean errors per rat for the 10 days was 2.8 errors for the experimental group, with a standard deviation of 1.1 errors. The mean errors per rat for the 10 days for the control group was 6.9 errors, with a standard deviation of 4.3 errors. Thus we can see from these results that there was a statistically significant difference between the learning rates of these two groups. The critical ratio of the difference between the means was 2.37, which enables us to reject the null hypothesis at the 2 percent level of confidence.

The change in the amount of choice point hesitation is shown by the VTE curves in Fig. 7. As these curves show

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50. Half of the experimental group ran west on all trials and half ran east. The control group was treated similarly. Thus the animals in both groups had on any given night to run either RLLR or LRRL depending on whether they were in the east or west-going group.

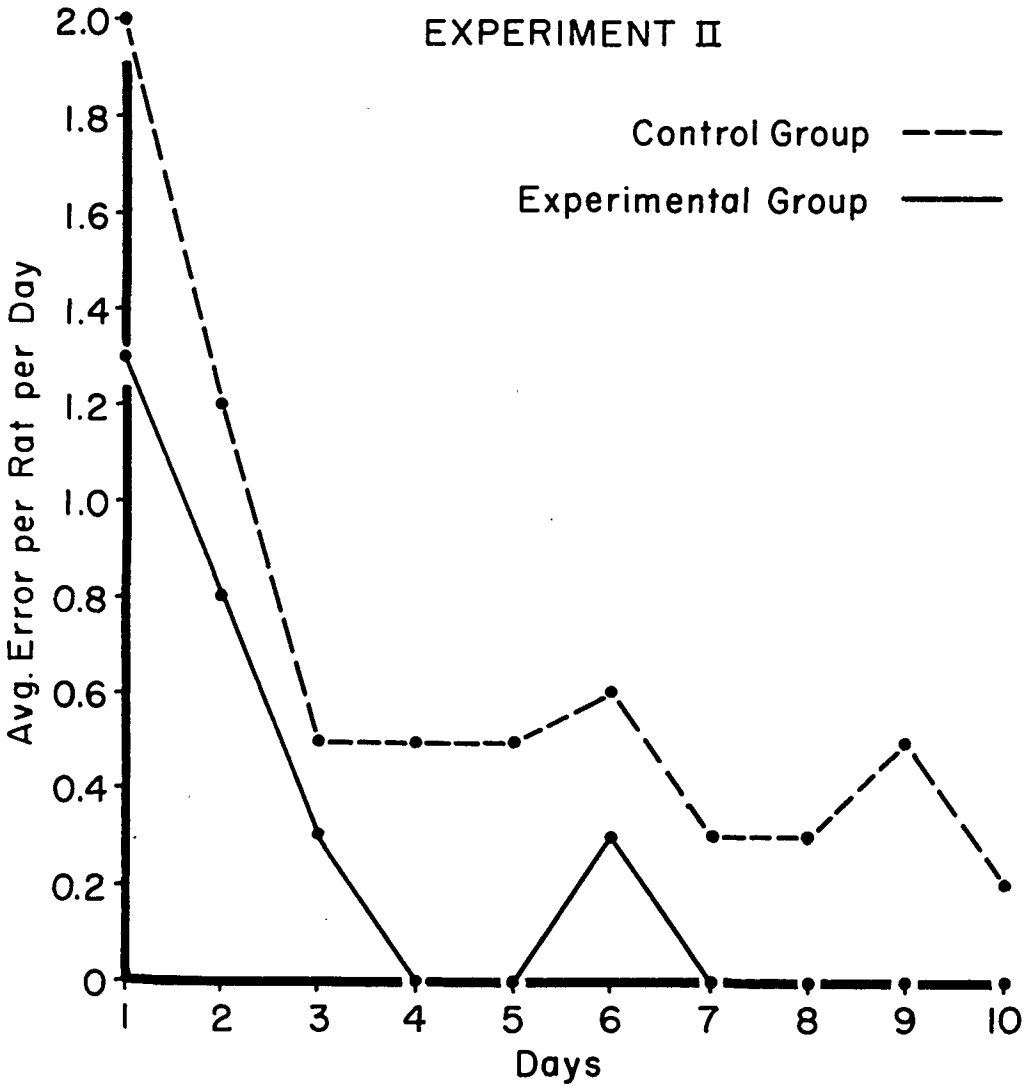

EXPERIMENT II
Control Group
Experimental Group
Avg. Error per Rat per Day
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
1
2
3
4
5
6
7
8
9
10
Days

Figure 6

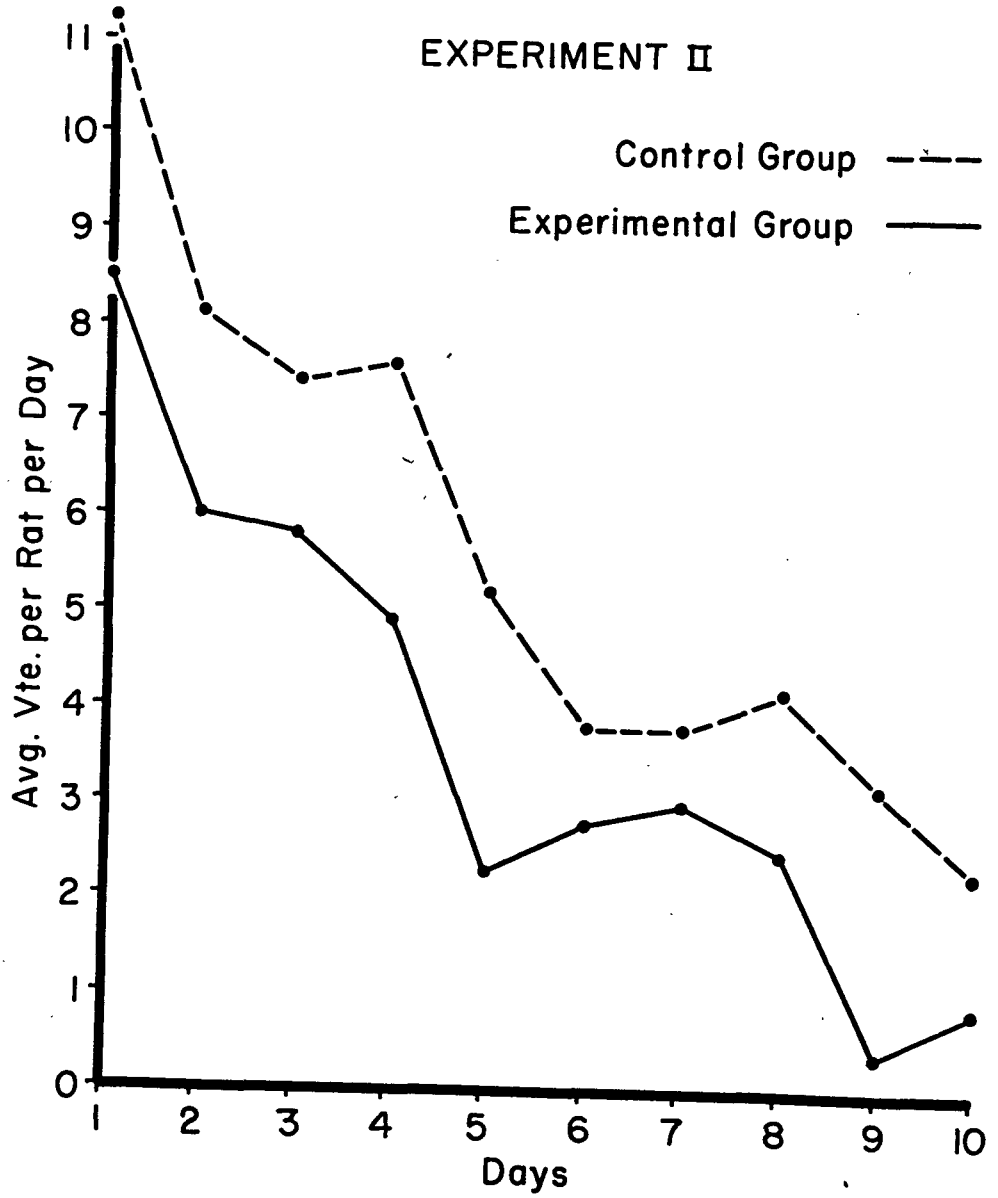

EXPERIMENT II
Control Group
Experimental Group
Avg. Vte. per Rat per Day
0
1
2
3
4
5
6
7
8
9
10
11
1
2
3
4
5
6
7
8
9
10
Days

Figure 7

the experimental group consistently exhibited less VTE's at the choice point than the control group. By the end of 10 days training, the experimental group was exhibiting less than 1 VTE per rat per day, while the control groups was exhibiting more than 2 VTE's per rat.

5. Discussion

The results of this second experiment indicate that the problem for the control or two-place group was more difficult than that for the experimental one-place group. The controls on the average made twice as many errors in the 40 trials and consitently exhibited more VTE's throughout the 40 trials.

Such results are consistent with the assumptions we made on pages 63-64 in explaining the results of the two-place versus one-place experiment. These assumptions implied (1) that when rats are trained as were the experimental group, the training from S to F will increase the disposition to turn correctly at the choice point when started at S ; and (2) that when rats are trained as were the controls, the training from S to F will not increase the disposition to turn correctly when started from S . In this sense, then, these assumptions implied that the experimentals would learn more rapidly than the controls.

Had the results been different,[51] we would, of course,

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51. This is the supposition that the experimentals do not learn more rapidly than the controls, and that S_2 is distinguishable from S_3 and F_1 from F_3.

have had to abandon our explanation of the two-place versus one-place experiment, for if any theorem derived from these assumptions is false, at least one of these assumptions must be false. It is in this sense that my experiment is an indirect test of these assumptions.[52]

Now although these results are consistent with my assumptions they may be consistent with other assumptions as well. I want now to consider what other kinds of assumptions would also explain the results of the present experiment and the original two-place versus one-place experiment. Now any proposed explanation of these results must account for the transfer of training which occurs when rats are t trained from S_1 and S_2 to F_1. That is, if any explanation is to be adequate, it must assume that some kind of disposition from S_2 is affected by training from S_1. My explanation, for example, assumes that the disposition which is affected is the shortest path disposition from S_2. Other explanations will differ from mine in assuming that some other kind of disposition is the crucial one.

Thus the question of greatest interest is that of determining what kinds of dispositions from S_2 are produced

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52. The reader may have wondered why I failed to test the assumptions directly. The answer is that, although they are logically prior to the theorems, they only occurred to me after I had done the experiments which test the theorems.

when rats are trained to run to F_1 from S_1. The reader will recall that my assuptions implied the following theorem :

19.0 $$(x)P(x) \supset T(x)$$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to take the shortest path from S_2 to F_1.

Experiment III will test this theorem. Now this experiment is an extremely important one to perform since it will not only enable us to discover whether this theorem is true, but it will also enable us to discover, if this theorem is false, what dispositions, if any, are acquired from S_2 as a result of training from S_1.

EXPERIMENT III

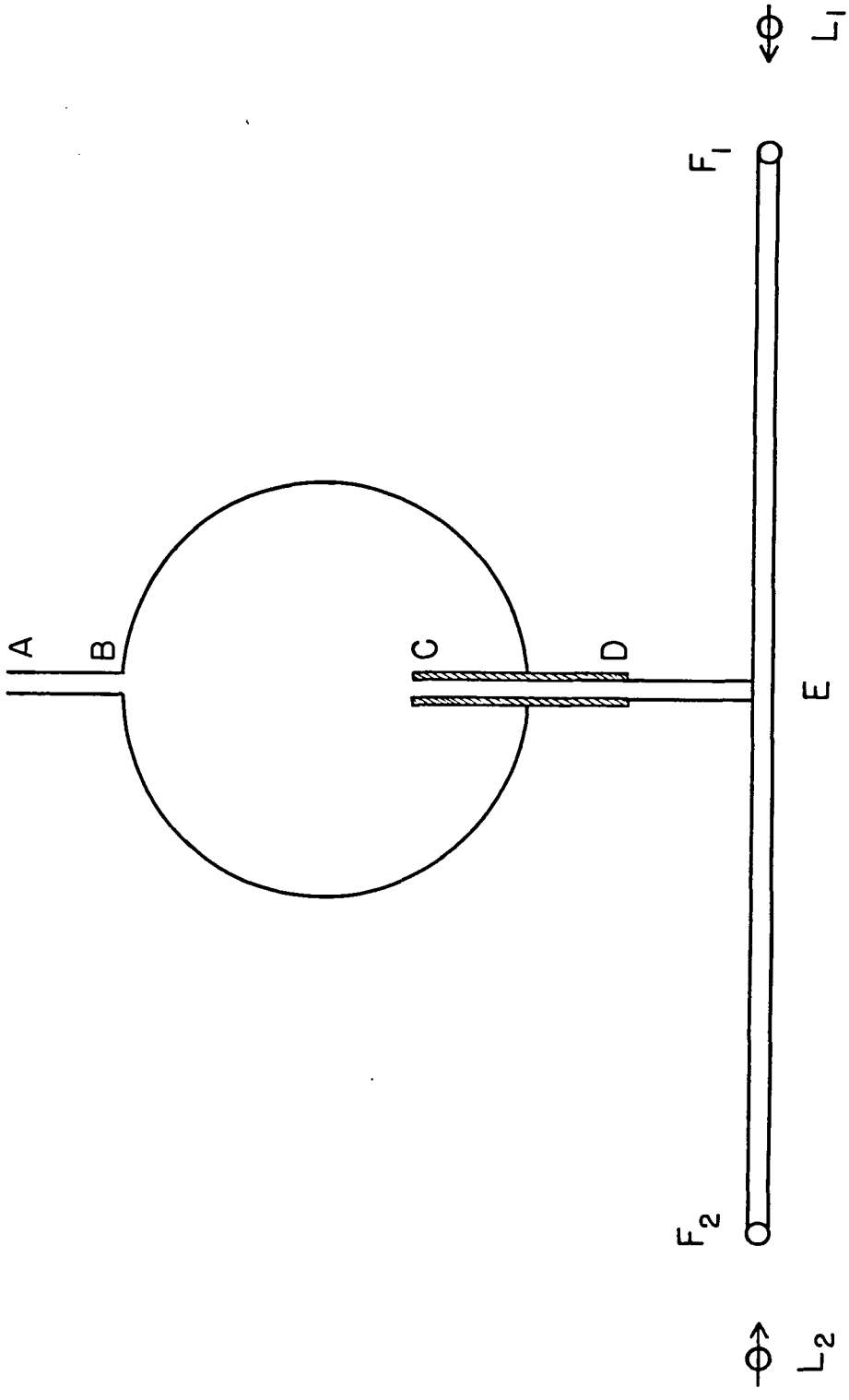
1. Animals

Fifty pigmented male rats, approximately 3 monts old were used in this experiment. These rats were M x M's, drawn from the Tryon stock in the rat colony at the University of California. None of them had had any previous maze training. They were all put on a 24 hour wet-food maintenance schedule 5 days before the beginning of the preliminary training. During the training they were run between eight

and eleven P.M.

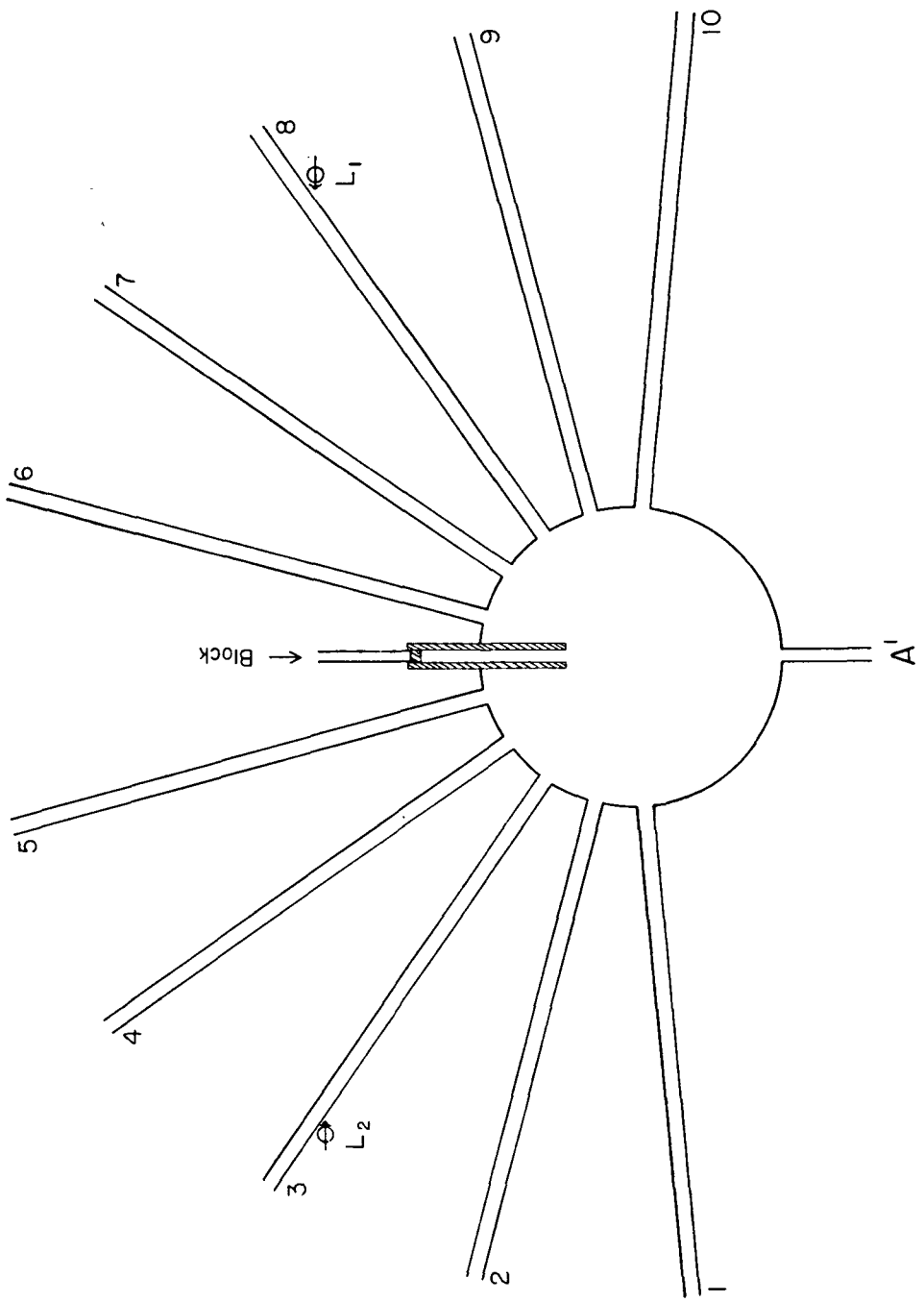
2. Apparatus

Figs. 8 and 9 present diagrams of the apparatus which were used. In Fig. 8 we see the apparatus used in the preliminary training. It consisted of an unpainted wooden circular table top which was 91 cms in diameter and several elevated wooden paths painted black. The path AB was 27 cms in length and was used as a starting path. The path CE was 75 cms in length, while path F_1F_2 was 2.5 meters long. At each end of the F_1F_2 path were small paper food cups 3 cms in diameter and 3 cms deep. Twenty cms beyond each end of the F_1F_2 path were small lights, L_1 and L_2, held in a socket so that the beam of light was directed on the food cup below. These lights were 15 watt bulbs, covered with paper cartons in which they were sold. They were 20 cms above the level of the F_1F_2 path. The only other illumination in the room was located at L_3 (see Fig. 10A). At this place was an ordinary desk lamp containing a 75 watt bulb. The reflector of this lamp was turned in such a way that its light was directed on the white wall of the room at an angle of 45 degrees. Thus this corner of the room (see Fig. 10A) was brightly lighted while the rest of the room was in deep shadow. Fastened to the sides of path DE were two pieces of unpainted plywood, which were 47 cms



Apparatus Used In Preliminary Training

Figure 8



Apparatus Used In The Test Trial

Figure 9

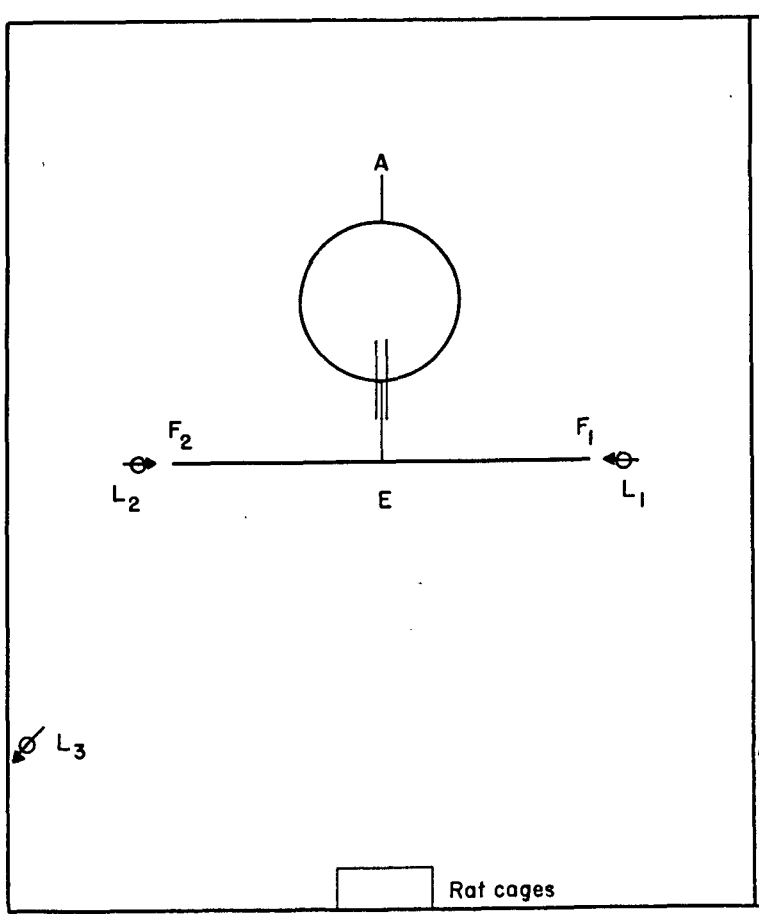


Figure 10A

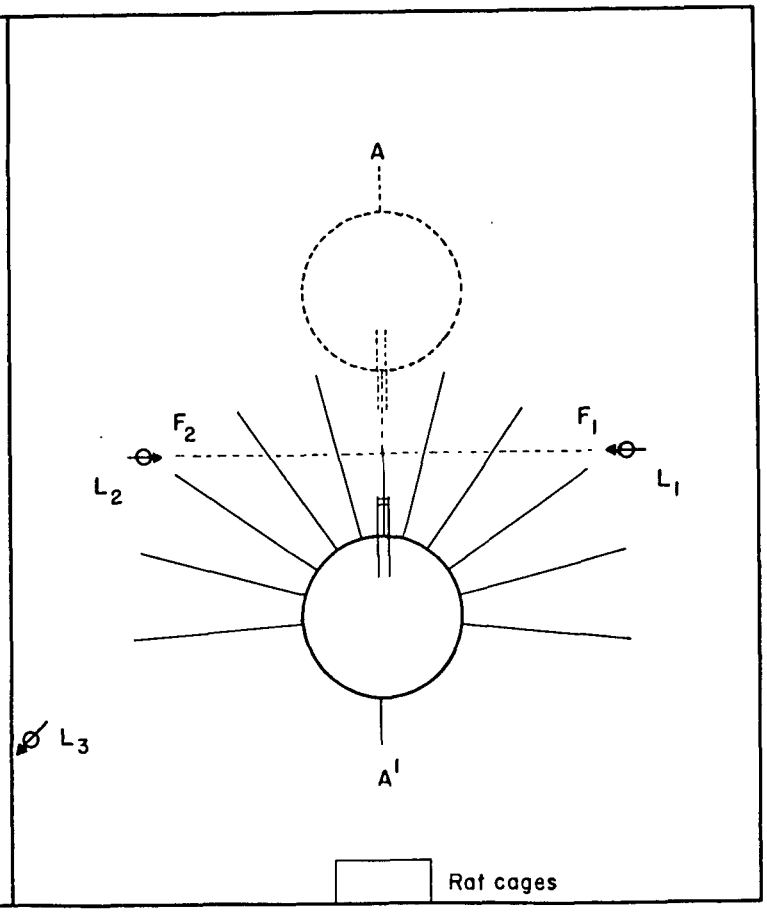


Figure 10B

Floor Plans Showing Positions of Apparatus.

Figure 10

in height and 47 cms in length. These formed an alley which began 20 cms from the center of the table top and ended 27 cms away from the choice point E.

In Fig. 9 we see the apparatus used in the test trial. This consisted of the same starting path, circular table top, alley on path CE, and lights at L_1, L_2, and L_3. But path F_1F_2 with the food cups was removed, and a block was placed at the end of the alley on path CD. Then 10 paths 1.5 meters in length were placed around the table top. These paths began at a point 95 degrees to the right of path CE and radiated in a counterclockwise fashion, each path being placed 20 degrees to the left of its neighbor.[53]

As can be seen in Fig. 10A the choice point E of the maze used in the preliminary training was in the center of the room and the F_1F_2 path ran parallel to the north and south walls. The apparatus used in the test trial was set exactly opposite to the other apparatus (see Fig. 10B) so that the starting point Á was now the same distance from the south wall as it had been from the north wall in the preliminary training. The position of the lights L_1, L_2, and L_3 remained, however, unchanged.

The rats lived in cages in another room and were carried in their cages from there to the experimental room

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53. There is an exception to this. Path 5 was 30 degrees to the left of path 6.

for their training. Their cages were put in a rack on the same axis as that of the starting path and choice point of the maze used in the preliminary training (see Fig. 10). On the test trial they were put in the same rack. The experimenter carried the rats by hand, put them on the apparatus by hand, and then sat on a stool beside the rack where the rats were kept.

3. Method

Pre-test procedures : On Days 1 and 2 the rats were trained in another room to accustom them to running on straight paths and to eating from paper food cups.

On Day 3 a block was placed at D on the apparatus in Fig. 8, and the rats were given 6 trials of running from F_2 to F_1.[54] On each of these trials the rats were allowed to eat at F_1 for 20 seconds and then were returned to the cage in the rack.

On Day 4 the rats were given 2 trials from D to F_1 (with a block at D) and two more trials from D to F_1 (with-

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54. Twenty-three of the rats were trained to find food at F_1. This group will be called "the F_1 group". Twenty-seven of the rats were trained to find food at F_2, and they were called "the F_2 group". In the following description of the training procedure we will consider only the F_1 group. The same kind of training was, of course, given the F_2 group.

out the block at D). Then they were given 2 trials through the alley from C to F .

On Day 5 they were given 2 more trials through the alley from C to F_1. Then they were given 4 trials from the start A to F_1.

On Days 6 and 7 they were given 8 daily trials from A to F_1. Thus after the training on Day 7, each rat had run 20 times from A to F_1.

Test procedures : On Day 8 one test trial was given. The apparatus was changed to that represented in Fig. 9 and placed in the room as pictured in Fig. 10B. Each rat was started at Á and allowed to explore the blocked alley, the table-top, and various alternative paths which radiated from it. If any rat put both front feet on any path but did not run out more than 2 feet on the path,[55] this was recorded as a "partial choice". A rat's trial ended as soon as it had chosen one of the paths and had run out on it at least 4 feet. If any rat took more than 7 minutes to make such a choice it was removed.

4. Results

On the test trial 8 of the 50 rats were discarded be-

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55. It was discovered in the first short-cut experiment (23) that any rat that ran more than 2 feet out on a path ran all the way out. The same behabior was observed in this experiment.

cause they failed to make the choice after 7 minutes. After having explored the paths, table-top, all of these 8 rats returned to the center of the table and refused to move from there except to return either to the alley or the starting path.

Fig. 11 shows the distribution of choices made by the remaining rats in the F_1 group. Inspection of this distribution shows that only 10.5 percent of the rats in the F_1 group chose path 8, the one which points most directly towards F_1. The distribution of choices in the F_2 group (see Fig. 12) is similar; only 4.5 percent of the rats in this group chose path number 3, the one which points toward F_2.

While only a few rats chose the most direct path to the original food place, the most frequently chosen path by both groups, was the last path on the side of the table which faced the original food place. Thus 47 percent of the F_1 group chose path number 10, and 65 percent of the F_2 group chose path number 1.

Now since the problem set for the F_1 group was the mirror image of that set for the F_2 group, the choices of both groups can be combined as in Fig. 13, where A is the last choice on the side facing food (eg. in the F_1 group, path 10, and in the F_2, path 1), C is the most direct path to the former food place (eg. in the F_1 group, path 8; in the F_2 group, path 3) and so forth. When the results of

both groups are combined in this way we see that 57 percent of the 42 rats chose the last path on the food side, while the distribution of rats on the other paths never exceeded 11 percent.

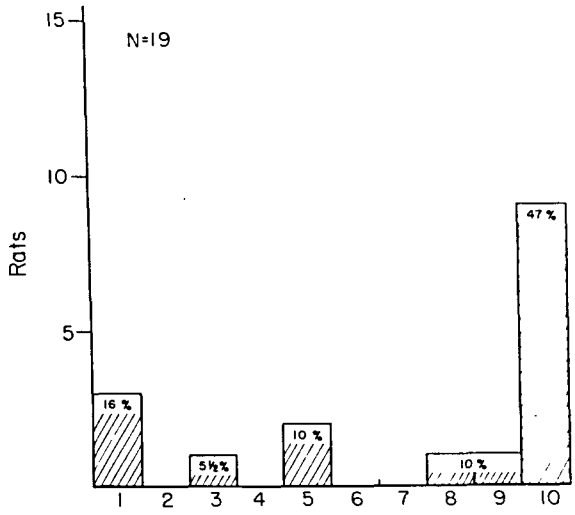
In Fig. 14 we see the distribution of partial choices (ie., in which the rats did not run out more than 2 feet) of the F_1 group, and in Fig. 15 we see the distribution of partial choices for the F_2 group. These two distributions show that the rats in both groups explored all paths about equally before making a choice. There was, however, one difference between the two groups. The F_1 group made almost twice as many partial choices as the F_2 group. The mean number of partial choices per rat for the F_1 group was 15.5 with a standard deviation of 8.5, while the mean number of partial choices per rat for the F_2 group was 7.2 with a standard deviation of 5.5.[56]

5. Discussion

The results of this third experiment indicate clearly that the rats did not acquire a disposition to run down the shortest path to the food location from S . As a consequence we must conclude that theorem 19.0 is false. This, of course, implies that one, and perhaps more, of the as-

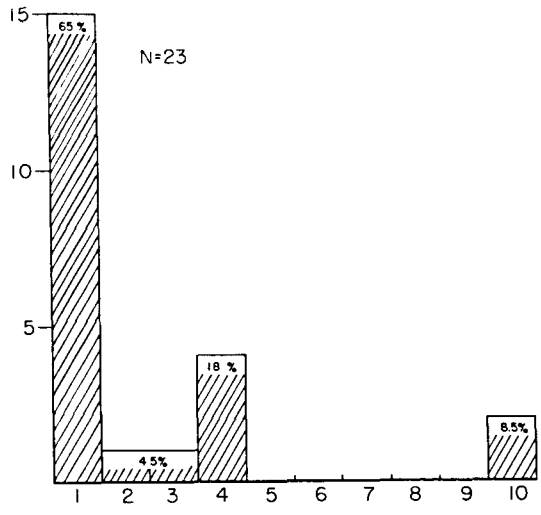
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56. The protocol records of all rats are given in Appendix III.



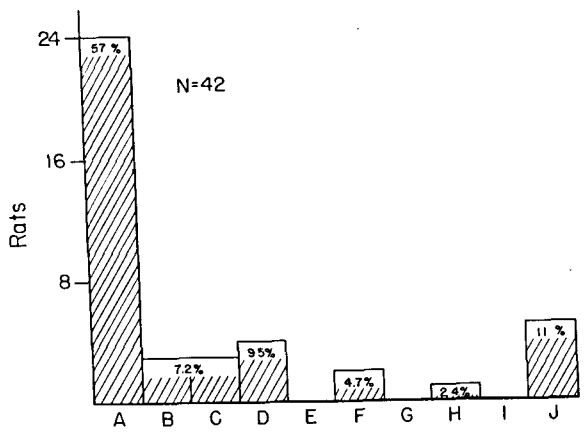
Paths Chosen by Rats in the F_1 Group

Figure 11



Paths Chosen by Rats in the F_2 Group

Figure 12



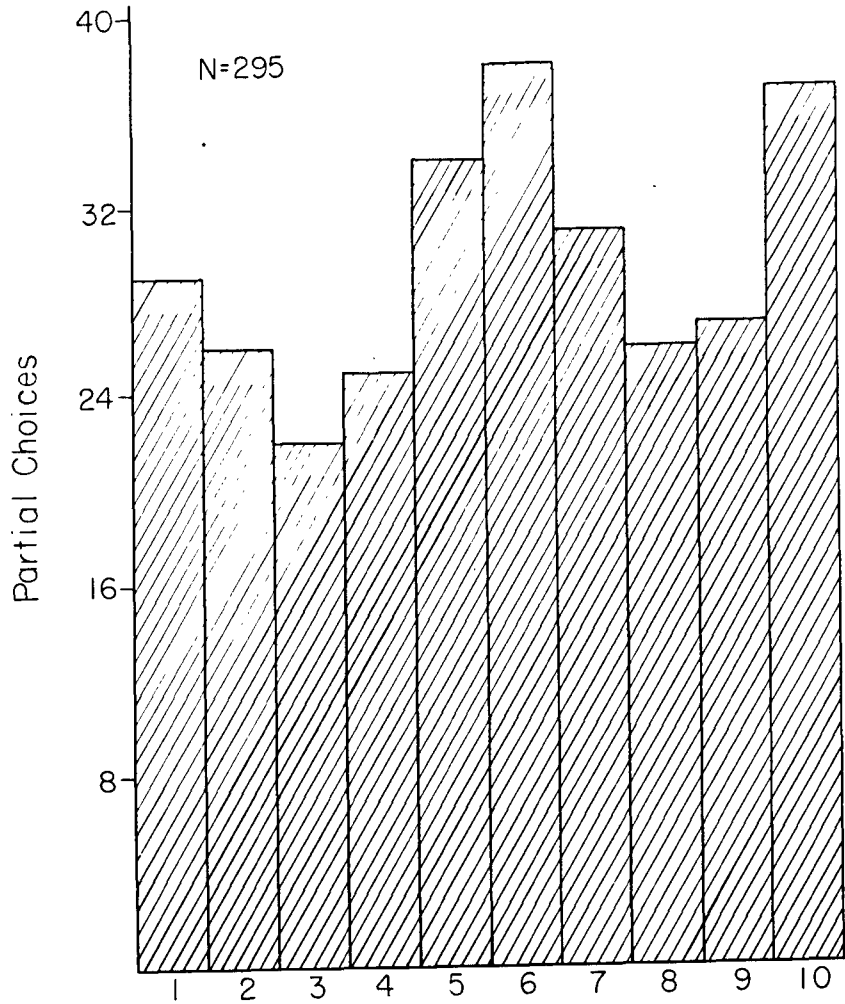
Paths Chosen by Rats in Both Groups

Figure 13

sumptions made in explaining the two-place versus one-place experiments is false. But before we consider the problem of revising these assumptions, I wish to discuss some other aspects of these experimental results.

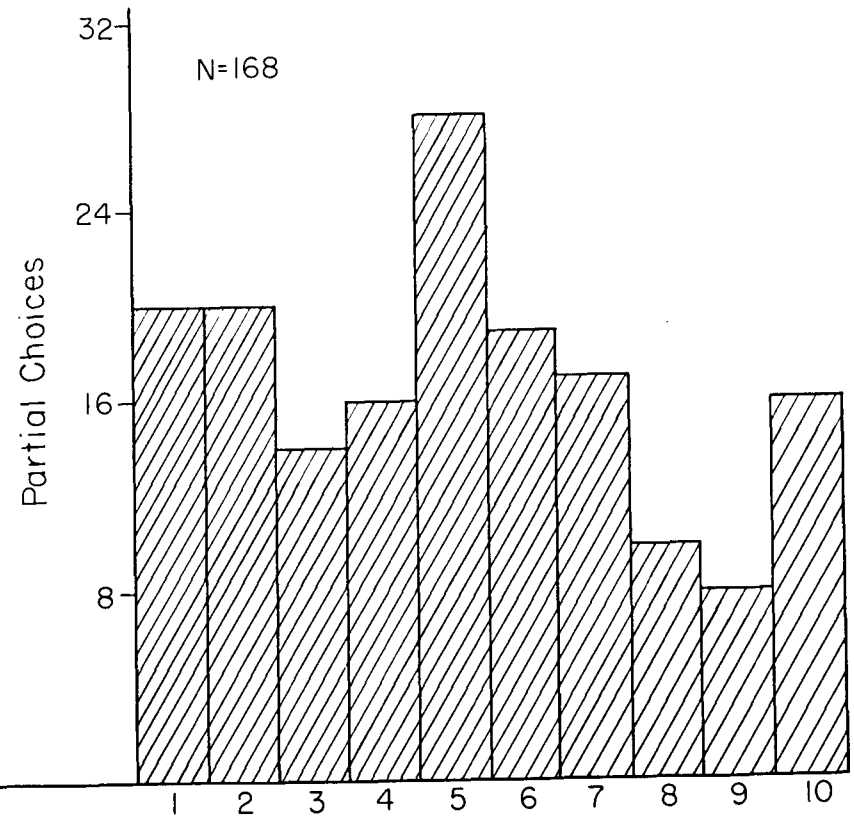
The fact that the rats in both groups tended to make partial choices of all the paths before making a final choice indicates that the final response or choice is not a stereotyped one. This is an important point since some critical readers may have supposed that the rats after coming out of the blocked alley made a stereotyped turn which took them to the path they finally chose. None of the animals in either group showed this kind of behavior, although 1 of the 42 rats ran from the start and chose a path without going into the alley or exploring any other path.

The fact that only 9.5 percent of the 42 rats chose lighted paths (see paths C and H in Fig. 13) is also important. The reader may remember that I argued on pages 53-54 that the disposition to take the practiced lighted path and the disposition to take a lighted path when the practiced one is blocked are not the same. This point is an important one for the interpretation of the results of the first short-cut experiment. The fact that only a few of the rats in the present short-cut experiment chose the lighted paths adds evidence to this point. Further evidence for this is also to be found in some experiments



Partial Choices in F_1 Group

Figure 14



Partial Choices in F_2 Group

Figure 15

which were run before the one which I have just described.

In these experiments a total of 56 rats were run under three different conditions in an attempt to discover whether they would take the shortest path from S_2 when trained from S_1. Of the 56 rats only one chose a lighted path. Let me describe these three preliminary experiments.

(1) Sixteen rats were run on a simple T-maze with the same F_1F_2 path as was used in the above experiment. The starting path was a straight piece of wood 4cms. wide and 50 cms. in length. The maze was placed in the room in the same way as the apparatus is placed in Fig. 10A with the choice point E in the center of the room. The distribution of lights was the same except that there was no light at L_3. After 20 trials on the simple T-maze, the rats were given a test trial on the apparatus shown in Fig. 9, placed in the room as indicated in Fig. 10B, except that there was no light at L_3. The distribution of choices showed no preference for any one path. There were very few partial choices and they tended to take the first path they found. None of them took a lighted path.

(2) Sixteen more rats were run on the same apparatus. The only difference between the training procedures was that in this new group, the rats were put on the circular tabletop (without the paths and without path CE) for 30 seconds after each trial on the T-maze. The test trial was the same

as that given the first group. Again the rats showed no preference for any one of the 10 paths and showed very few partial choices. One of the rats of this group chose a lighted path.

(3) Twenty-four rats were run on an apparatus like the one shown in Fig. 9 except that instead of having a choice at E the CE path continued straight ahead for 1.5 meters where there was a food cup with food. After the rats had learned to run across the table top and through the alley, they were trained on the simple T-maze used in the two preceding experiments. After 20 trials of this, they were given a test trial exactly like the ones given to the other two groups. In this case there was a lot of exploration and partial choices. But the paths which were most frequently chosen were paths 5 and 6, the ones which pointed most directly towards the food place at the end of the CE path used in the preliminary training. None of these rats chose a lighted path.

On the basis of all this evidence it seems clear that the disposition to take a lighted path when the practiced lighted path is blocked is not the same as the disposition to take the practiced lighted path.

I have pointed out that at least one of the assumptions in my explanation must be false if theorem 19.0 is false.

Now it seems quite plausible from our results that assumption 15.2 is false. This is the one which states that all rats which have the disposition to take the shortest path from S_1 to F_1 also have the disposition to take the shortest path from S_2 to F_1. If this is the case, do the results of our experiment suggest any way in which this assumption can be changed so that it may be true and still entail the theorems which explain the two-place versus one-place experiments? I think they do.

Let us assume that when rats are trained to run to F from S they also acquire a disposition to run in the same direction as the practiced path, say west.[57] Now if we replace 15.2 by the following assumption :

15.21 For all x, x has a disposition to run west from $S_1 \supset$ x has a disposition to run west from S_2

and make the appropriate changes in the other assumptions,[58] we should be able to derive theorems 16.2 and 17.1 which explain earlier Tolman-Ritchie-and-Kalish two-place versus one-place experiments, and theorems 16.0 and 18.0 which ex-

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57. This disposition will be defined in exactly the same way as the shortest path disposition (defined by 8.0) except that the second matrix will be "x runs down the path which points directly west."

58. We must substitute the direction disposition for the shortest path disposition.

plain Experiment II in this thesis. Furthermore, theorem 19.0 which the present experiment has shown to be false will not be entailed by these assumptions, whereas the following theorem, which is verified by the present experiment, will be entailed by these new assumptions :

19.1 For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to run west from S_2.

In conclusion then, the fact that my rats exhibited a direction disposition on the test trial suggests a way to revise the assumptions.

This does not of course mean that I am asserting that these new assumptions are true. We are still a long way from understanding simple T-maze learning, and the experiments which I have described are exploring new aspects of this field of investigation. We may discover later that both direction and shortest path dispositions are acquired, but that in some situations, such as the ones we considered, the direction disposition is so strong that it masks the shortest path disposition. More experiments of this kind will have to be performed before we will be able to state a set of assumptions which will be adequate. Thus the assumptions I have stated and the experiments I have performed are presented not because they are intended to be definitive but because they exemplify (1) what kinds of facts we ought

to try to discover, and (2) the way we should organize these facts, if we wish our investigations to proceed in an orderly fashion.

I have emphasized that at present we know very little about T-maze learning. This is an important thing to remember. But I do not believe that our ignorance is an excuse for loose thinking and vague language. The main benefit of verbal precision is that it enables us to tell when a statement is false, and, furthermore, it enables us to tell what other statements must be false if this one is false. When we are as ignorant as we are, we can only progress by ridding ourselves of the false notions which are implicit in our vague ideas. Only by using precise methods and clear statements will we be able to get rid of these false hypotheses.

PART III

CONCLUSIONS

GENERAL SUMMARY AND CONCLUSIONS

The present paper arose from an interest in maze learning and from a dissatisfaction with all the current theoretical notions used to describe and explain maze learning. Most of the current theories of behavior are expressed in a way that leads the reader to suppose that the subject matter is the relation between stimuli and responses, and that the laws for which such theoreticians search are the laws which relate such stimuli and responses. Such notions have always seemed to me to be no more than misleading metaphors. What such notions might mean if taken literally, I have never been able to understand or even to imagine. But if we turn from their theories and instead examine their experiments we discover that the relations for which they appear to be searching are not the relations between stimuli and responses, but are the relations between the training given an animal and the dispositions which the animal acquires as a result of this training.

After this discovery it seemed to me essential to find some simple and precise way of describing the results of such experiments. The complexities of the traditional S-R method of describing such results increased when it was applied to learning experiments where the stimuli presented to the animal were not controlled and the responses made

by the animal were not easily observed (eg. as in maze learning or in discrimination learning). This complexity was further increased by mathematical and logical elaboration of notions which were fundamentally vague and ambiguous. Such a use of mathematics and logic only obscures the issues, making it even more difficult to determine the content. As a consequence I wanted to start over again, using only the simplest notions of mathematic logic to clarify rather than to obscure the basic notions.

It seemed important to me therefore to take great pains to make clear the logical form of the definitions and laws which I was to employ. Although psychologists have concerned themselves with many abstruse philosophical problems, few of them have concerned themselves with this one. Most of them have been satisfied with affirming that they "believe in" operational definitions and with saying that their "terms are operationally defined." None of them, however, has given an example of an operational definition. Instead the usual procedure is to say that they wish to define some term operationally. They then describe an experiment and conclude by saying that this experiment defines the term.

In Carnap's paper "Testability and Meaning" (5) I found the logical form required for the kind of definitions which were needed to introduce the disposition predicates

which I was to use. In the beginning of the introduction to this paper I gave many examples of the logical and empirical problems which arise when we try to define disposition predicates and state laws relating these dispositions to various training procedures. In doing this I discovered that even for a simple disposition like the conditioned eyelid response habit, many tiresome details must be specified in our definition if it is to be precise. I further discovered that when stated in this way there was clearly no single principle of reinforcement which would take the place of the disjunction in law 11.1 (see page 21).

Now when we returned to consider S-R psychology it became clear that S-R psychology is based upon two assumptions and that both of them are false. The first assumption is that some particular principle of reinforcement (eg. substitution, or effect, etc.) states the essential factors common to all training procedures which produce conditioned responses. But law 11.1, as I have just remarked, shows that this assumption is false. The second assumption is that all habits or acquired dispositions are essentially similar. An analysis of the definitions for the conditioned eyelid response habit and for the conditioned salivary response habit show that the second assumption is false. The definition for the latter habit must contain the matrix "x is deprived of food etc." in the antecedent, whereas

this matrix does not appear in the definition of the eye-lid habit. We must conclude then that these two habits are not essentially similar and that the second assumption is false.

I then decided to revise the two assumptions of S-R psychology to see if they are true when restricted to simple T-maze learning. I defined the matrix "x learned the maze" as equivalent to the matrix "x has a disposition to run directly to F, when deprived and put on the maze" (see pages 40-42). The first assumption then becomes law 13.0 (see page 43), and the second assumption becomes the statement that the disposition to run directly to F etc. is the only kind of disposition acquired by this training. The results of the latent learning experiments demonstrate that the first assumption is false, while the discovery of the short-cut disposition (see pages 50 ff) shows that the second one is false.

The discovery of the short-cut disposition led to the Tolman, Ritchie and Kalish two-place versus one-place experiment. Having discovered that rats acquire other dispositions than the disposition to take the practiced path, they wondered whether rats would not learn to go to the same food place by two different paths more readily than they learned to go to two different food places. The results of this experiment showed conclusively that this guess

was correct. The present writer then suspected that the short-cut disposition, or some similar disposition, was the crucial one which accounted for these results. As a consequence he tried to state an explanation of these results in terms of the relation of the short-cut disposition to the disposition to make, say, right turns at the choice point, when started at, say S_1. This explanation consists of four empirical assumptions which entail the results which were confirmed in the Tolman, Ritchie and Kalish two-place versus one-place experiment.

The first theorem is : (1) A group of rats trained from S_1 to F_1 and S_2 to F_1 will learn faster than a group trained from S_1 to F_1 and S_3 to F_3 .

The second theorem is : (2) A group of rats trained from S_1 to F_1 will have a disposition to take the shortest path from S_2 to F_1.

The first two of the following experiments were designed to test the first theorem.

In Experiment I two groups of rats were trained on comparable single unit maze problems. The experimental group was required to learn to go from two starting paths to the same food place, while the control group was to learn to go from two paths to two respectively different food places. No significant difference in the learning rates of the two groups was discovered. The reasons for these

results were investigated and it was discovered that the orientational cues had not been adequately controlled to test the theorem.

In Experiment II the orientational cues were better controlled and two new groups of rats were run in the same type of problem. At the end of 10 days for Experiment II the mean error per rat in the control group was 6.9 errors with a standard deviation of 4.3 errors. The mean error per rat in the experimental group was, however, only 2.8 errors with a standard deviation of 1.1 errors. The critical ratio of the difference between the means was 2.37. As a consequence of these results I concluded that the first theorem was probably true.

To determine whether the second theorem was true Experiment III was performed. Fifty male rats were given 20 trials in a single unit T-maze from S_1, and then tested with the short-cut apparatus from S_2. Since only 7.1 percent of the rats which made final choices chose the shortest path to the original food place, I concluded that the second theorem was probably false.

Since 57 percent of the rats in Experiment III chose the last path on the food side, I suggested that the assumptions be revised so that the direction disposition (eg. to run west) replace the shortest path disposition. This revision of the assumptions enables us to derive the theorems

which explain both two-place versus one-place experiments without entailing any theorems which at present are known to be false.

PART IV

APPENDICES

APPENDIX I

Below are the definitions and laws cited throughout this paper. They are listed in the same order as they appear. After each one is the page on which this expression first appears. All definitions are designated by numbers in the range from 1.0 to 9.9, while all synthetic sentences or laws are designated by numbers in the range from 10.0 to 19.9.

1.0 x has the disposition D $\equiv$ x is in water $\supset$ x dissolves. (p. 5)

2.0 $P(x) \supset (Q(x) \equiv R(x))$ (p. 6)

2.1 x is in water $\supset$ (x dissolves $\equiv$ x has a disposition to dissolve in water). (p. 6)

2.12 x is soluble $\equiv$ x has a disposition to dissolve in water. (p. 6)

2.2 x is exposed to air $\supset$ (x evaporates $\equiv$ x has a disposition to evaporate when exposed to air). (p. 7)

2.21 x is volatile $\equiv$ x has a disposition to evaporate when exposed to air. (p. 8)

10.0 For all x, x is hot and the pressure of the air surrounding x is low $\equiv$ x is volatile. (p. 8)

3.0 x is presented with a light $\supset$ (x makes eyelid response $\equiv$ x has a disposition to make an eyelid response when a light is presented). (p. 12)

3.1 x is presented with light at t $\supset$ (x makes an eyelid response after t + θ and before t + e $\equiv$ x has a disposition to make an eyelid response when a light is presented). (p. 14)

3.12 x is presented with light at t. No airpuff is presented between t and t + e $\supset$ (x makes an eyelid response after t + θ and before t + e $\equiv$ x has a disposition to make an eyelid response when a light is presented and no airpuff is presented). (p. 14)

3.2 x is presented with light at t. No airpuff and no distracting stimulus is presented between t and t + e $\supset$ (x makes an eyelid response after t + θ and before t + e $\equiv$ x has a disposition to make an eyelid response when a light is presented and no airpuff or distracting stimulus is presented). (p. 15)

3.21 x has a conditioned eyelid response habit to light $\equiv$ x has a disposition to make an eyelid response when a light is presented and no airpuff or distracting stimulus is presented. (p. 16)

11.0 For all x and all t, x is given N trials, in each of which a light is presented at t and an airpuff at t + e $\equiv$ x has a conditioned eyelid response habit to light. (p. 16)

11.1 $(x)P(x) \vee H(x) \vee A(x) \vee B(x) \equiv Q(x)$

For all x, x is given training P or H or A or B $\equiv$ x has a conditioned eyelid response habit to light. (p. 20)

4.0 x is given training B and is now presented with light $\supset$ (x makes an eyelid response $\equiv$ x has a conditioned eyelid response to light) (p. 24)

5.0 x is presented with tuning fork at t and no food and no distracting stimulus is presented between t and t + e $\supset$ (x salivates after t + θ and before t + e $\equiv$ x has a conditioned salivary response habit to a tuning fork). (p. 36)

12.0 For all x and all t, x is given N trials, in each of which x is deprived of food and a tuning fork is presented at t and dry food at t + e $\supset$ x has a conditioned salivary response habit to a tuning fork. (p. 37)

5.1 x is deprived of food for N hours before t and is presented with a tuning fork at t and no food and no distracting stimulus is presented between t and t + e $\supset$ (x salivated after t + θ and before t + e $\equiv$ x has a conditioned salivary response habit to a tuning fork). (p. 38)

6.0 x is deprived of food for N hours before t, and x is on start of maze at t and eats no food between t and t + e $\supset$ (x runs directly to F, avoids B, and arrives at F before t + e $\equiv$ x has a disposition to run directly to F, when deprived and put on the maze). (p. 40)

6.1 x has learned the maze $\equiv$ x has a disposition to run directly to F, when deprived and put on the maze. (p. 42)

13.0 For all x and all t, x is given N trials in each of which x is deprived of food N hours before t, x is put on the start of the maze at t, x goes over the choice point, and x eats for N seconds at F $\equiv$ x has learned the maze. (p. 43)

7.0 x is deprived of food and x has been trained on path P and x is now put on path P and P is blocked and there are other paths which lead away from path P, one of which points directly to location L $\supset$ (x runs down the path which points directly to location L $\equiv$ x expects food at location L). (p. 54)

14.0 For all x, x has training $T_1 \equiv$ (x has D_1 and x has D_2 and x has D_3) v x has D_4. (p. 58)

7.1 x is deprived of food for N hours before t and x is on path at t and x runs directly into alley before t + e and path P is blocked, and there are other paths which lead away from path P, one of which points directly to location L $\supset$ (x runs down the path which points directly to location L $\equiv$ x has a disposition to short-cut to L when deprived, when the practiced path is blocked and when x runs into the block). (p. 58)

8.0 x is deprived of food for N hours before t and x is on path at t and there are paths which lead away from path P, one of which points directly to $F_1 \supset$ (x runs down the path which points directly to $F_1 \equiv$ x has a disposition to take the shortest path to F_1 when deprived). (p. 62)

15.0 $(x)P(x) \supset Q(x)$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to short-cut from S_1 to F_1. (p. 63)

15.01 $(x)Q(x) \supset R(x)$

For all x, x has a disposition to short-cut from S_1 to $F_1 \supset$ x has a disposition to take the shortest path from S_1 to F_1. (p. 63)

15.1 $(x)R(x) \supset S(x)$

For all x, x has a disposition to take the shortest path from S_1 to $F_1 \supset$ x has a disposition to turn right at C from S_1. (p. 64)

15.2 $(x)R(x) \supset T(x)$

For all x, x has a disposition to take the shortest path from S_1 to $F_1 \supset$ x has a disposition to take the shortest path from S_2 to F_1. (p. 64)

15.3 $(x)T(x) \supset U(x)$

For all x, x has a disposition to take the shortest path from S_2 to $F_1 \supset$ x has a disposition to turn left at C from S_2. (p. 64)

16.0 $(x)P(x) \supset S(x).U(x)$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to turn right at C from S_1 and left at C from S_2. (p. 65)

16.1 $(x)\acute{P}(x) \supset S(x).U(x)$

Where "$\acute{P}(x)$" means "x is trained from S_2 over choice point C to F_1." (p. 65)

16.2 $(x)\acute{P}(x).P(x) \supset S(x).U(x)$ (p. 65)

17.0 $(x)L(x) \supset M(x).N(x)$

For all x, x is trained from S_2 over choice point C to $F_2 \supset$ x has a disposition to turn right at C from S_2 and left at C from S_1. (p. 66)

17.1 $(x)P(x).L(x) \supset M(x).N(x).S(x).U(x)$ (p. 66)

18.0 For all x, x is trained from S_3 over choice point D to $F_3 \supset$ x has a disposition to turn left at D from S_3 and right at D from S_4. (p. 68)

19.0 $(x)P(x) \supset T(x)$

For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to take the shortest path from S_2 to F_1. (p. 69)

15.21 For all x, x has a disposition to run west from $S_1 \supset$ x has a disposition to run west from S_2. (p. 109)

19.1 For all x, x is trained from S_1 over choice point C to $F_1 \supset$ x has a disposition to run west from S_2. (p. 110)

APPENDIX II

The following remarks are intended to acquaint the reader with the meaning of the logical notation which I have employed throughout this paper.

A simple sentence is a linguistic structure of the form "S is P", where "S" is the name of the subject and "P" is the name of the predicate. We will use the letters "p" and "q" etc., as variables taking nothing but simples sentences as their values.

Any sentence formed of simple sentences by means of one or more connectives is a molecular sentence. Connectives are expressions such as "either...or..", and "and". These two binary connectives will be represented by the symbols "v" and "." respectively. Thus any sentence of the form "pvq" or "p.q" is a molecular sentence.

A mode of sentence composition will be called a "truth function" if the truth-value of the compound is determined solely and in all cases by the truth-value of the components. Simple sentences cannot be truth-functions since they are not composite. All the molecular sentences with which we will be concerned are truth-functions. Thus, the truth-value of any molecular sentence of the form "pvq" is determined completely by the truth values of p and q. In other words if we are to determine the truth-value of any

sentence of this form we must know the truth-values of P and q.

Consider now the following example of a sentence of this form.

This is black v that is white

Now since p has two possible truth-values (ie. either T or F) and q also has two possible truth values, there are four possible different combinations of the truth-values of p and q together. These four possible combinations are represented in the table below.

	p	q
1	T	T
2	F	F
3	T	F
4	F	T

table 1.

Now for certain of these combinations the molecular sentence "This is black v that is white" is true. Which are these combinations? They are combinations 1, 3 and 4 in table 1. That is, this molecular sentence is false, only when this is not black and that is not white. In all other cases it is true.

Consider now the following example of a sentence of the form "p.q"

This is hot.that is cold.

Now what are the combinations of truth-values of p and q represented in table 1 for which this sentence is true? There is only one such combination, namely 1. That is, this molecular sentence is true only when this is hot and that is cold. In all other cases it is false.

Now if we add to table 1. a third column and place the name of some molecular sentence form (eg. 'pvq') at the head of this third column, we can represent the truth values of all sentences of this form parallel to each of the combinations of truth-values of their components. Thus this new third column is called "the truth-table" of sentences of that molecular form. Table 2 contains in the third column the truth-table for 'pvq'.

	p	q	'pvq'
1	T	T	T
2	F	F	F
3	T	F	T
4	F	T	T

table 2.

and table 3 contains the truth-table for "p.q"

	p	q	'p.q'
1	T	T	T
2	F	F	F
3	T	F	F
4	F	T	F

table 3.

In our preceding discussion we have determined the truth-values of molecular sentences for the various combina-

tions of the components by examining common usage, and have represented the results of this examination diagrammatically by means of a truth-table. However, the truth-table has a much more significant use than that of expressing clearly the results of analysis. It provides a means by which sentence connectives can be defined precisely. The truth-table provides a means of defining a connective because it states the truth value of a statement formed by the connective for the various combinations of truth-values of the components : that is, it states under what conditions a statement formed by this connective is true, and under what conditions false. When we have stated the conditions which determine the truth or falsity of a statement containing a connective we have stated the meaning of this connective.

Let us define the connective, "⊃", in such a way that the molecular sentences formed by it are true whenever the sentence preceding the connective is false or the sentence following the connective is true. The sentence preceding the connective is called the "antecedent" and the other is called the "consequent". It follows, then, that molecular sentences formed by this connective are false only if the antecedent is true and the consequent is false. This decision is represented in table 4.

	p	q	'p⊃q'
1	T	T	T
2	F	F	T
3	T	F	F
4	F	T	T

table 4.

A molecular sentence of the form "p⊃ q" is read "if.. then...". This connective is called by logicians "material implication" and can be used to express any of the ordinary uses of the conditional. The important point to note is that table 4 is not meant to be a truthful representation of how people use sentences of the form "if p then q", but a representation of our decision to use sentences of this kind in this way. Although the truth tables of "pvq" and "p.q" were not presented as representations of decisions, we will, from this point on, consider them as decisions, that is, as definitions of the connectives "v" and ".".

Another connective which logicians find useful and of which we will make use is "≡". This connective is read "is equivalent to", and is defined as follows :

	p	q	'p ≡ q'
1	T	T	T
2	F	F	T
3	T	F	F
4	F	T	F

table 5.

Table 5 tells us that "p ≡ q" is true only if the truth values of the components are the same; if the components differ in truth value, then "p ≡ q" is false.

One other connective will be defined. It differs from those we have discussed by being a singularly connective rather than a binary connective, that is, it is a method of

elaborating a single sentence to form a new sentence. This connective is symbolized "-", and is read "not" or "it is false that". Molecular sentences formed by this connective have only one component; the molecular sentence always differs in truth-value from its component. This decision is represented in table 6.

	p	-q
1	T	F
2	F	T

table 6.

Truth-tables have a second use. Also, by means of them we can give a precise meaning to the term "analytic" (ie. "logically true").

A sentence is analytic if its truth-table contains no letter other than "T". Let us examine the truth-table for all sentences of the form "pv-p". This form is commonly called "the law of excluded middle."

	p	-p	'pv-p'
1	T	F	T
2	F	T	T

table 7.

Column 1 of table 7 contains the possible truth-values of one of the components of 'pv-p'; column 2 contains the possible truth-values of the other component. Column 3 contains the truth-values of 'pv-p' for all the possible truth-values of its components. This column contains no letter

other than "T", hence 'pv-p' is analytic.

One more example should suffice to illustrate the application of the definition of "analytic". Let us examine the truth-table of sentences of the following form : "p ⊃ (q ⊃ p)".

	q	p	'q ⊃ p'	'p ⊃ (q ⊃ p)'
1	T	T	T	T
2	F	F	T	T
3	F	T	T	T
4	T	F	F	T

table 8

Columns 1 and 2 contain the possible combinations of the truth-values of the simple components; column 3 contains the truth-values of the molecular component 'q ⊃ p', and column 4 contains the truth-values of 'p ⊃ (q ⊃ p)'. Column 4 contains no letter other than "T", hence "p ⊃ (q ⊃ p)" is analytic.

Thus, we have illustrated the meaning of the decision to call any sentence of a given molecular form "analytic" if its truth-table contains no letter other than "T". This decision can be expressed negatively by stating that it is impossible to assign truth-values to the components of an analytic statement which would make the truth-value of the statement falsehood. This is what is meant when one says that no empirical discovery can affect the truth of an analytic sentence; that is, nothing that we discover or fail to

discover concerning the truth of the components can affect the truth of the composite sentence.

An explicit definition is a declaration that a newly introduced symbol or combination of symbols is to mean the same as a certain other combination of symbols of which the meaning is already known. We will give the names "definiendum" and "definiens" respectively to what is defined and to that which it is defined as meaning.[1] To say that one symbol has the same meaning as another is to say that it is a decision that whatever is grounds for affirming or denying a statement in which one of the two symbols occurs is also the grounds for affirming or denying the statement when this symbol is replaced by the other.

The equivalence sign, "≡", immediately suggests itself as a means of expressing what we are calling an "explicit definition" because the expressions connected by it are asserted to have the same truth-values. Now, since the equivalence sign is a sentence-connective, the types of expressions connected by it are sentences or expressions capable of becoming sentences, namely, matrices. Hence, if we use the equivalence sign to state an explicit definition, we must introduce our new symbol by means of a matrix equated to a matrix containing the already defined symbol. The former is our definiendum and the latter our definiens. Our declaration is to the effect that these two matrices

are formally equivalent; that is, we decide that they are equivalent for all values of the variable.

Thus, by an explicit definition of a description predicate "Q" with one subject we understand an expression of the form :

$$Q(x) \equiv \ldots x \ldots$$

where at the place of "...x..." stands a matrix, the definiens, which contains "x" as its only free variable and a finite number of previously explicitly or ostensively defined descriptive predicates.

In an explicit definition, the equivalence sign is used to express a decision that two terms have the same meaning; this decision is expressed by asserting that matrices containing the terms are formally equivalent. And it follows from the nature of formal equivalence plus the decision that each can replace the other in any truth-function that contains either, without the possibility of affecting the truth-value of the truth-function. However, the equivalence sign is also used to express a discovery that two sentences as a matter-of-fact have never been found to differ in truth-value. The equivalence in this case, even if expressed as a formal equivalence, for instance :

(P-law) For all x, x dissociates into hydrogen and oxygen $\equiv$ the density of x is less when solid than when liquid.

does not warrant an assertion to the effect that either member of the equivalence can replace the other in any truth-

function without the possibility of effecting the truth-value of the truth-function. The distinction being referred to, namely, the distinction between an equivalence which expresses a decision and one that expresses a statement of fact, is of paramount importance.

A conditioned definition is a declaration that, provided that a certain condition is fulfilled, a newly introduced symbol or combination of symbols is to mean the same thing as a certain other combination of symbols of which the meaning is already known. If the condition is not fulfilled the two sets of symbols do not have the same meaning. Let us restate this in the following manner : A conditioned definition, like an explicit definition, contains a new symbol and an already defined one. Now a conditioned definition is a declaration that we will consider that whatever is evidence for a statement containing the already defined symbol is also evidence for a statement containing the newly introduced symbol provided that a certain condition is fulfilled. If the condition is not fulfilled the evidence for the statement containing the already defined symbol is not evidence for the statement containing the newly introduced symbol.

Thus, a conditioned definition consists of three components : 1) a matrix describing a condition; 2) a matrix containing an already defined symbol; and 3) a matrix con-

taining a new symbol. If a value is assigned to the free variable of these matrices, then :

I. If (1) is true, the evidence for (2) can be taken as evidence for (3).

II. If (1) is false, the evidence for (2) can not be taken as evidence for (3).

As Carnap has shown the molecular sentence form capable of satisfying these conditions is :

$$p \supset (q \equiv r)$$

where "p", "q", and "r" are replaced by (1), (2), and (3) respectively. That this form expresses exactly what is intended by a conditioned definition can be most readily recognized by an examination of its truth-table.

	1	2	3	4	5
	q	r	p	'q ≡ r'	'p ⊃ (q ≡ r)'
1	T	T	T	T	T
2	F	T	F	F	T
3	F	T	T	F	F
4	T	T	F	T	T
5	T	F	T	F	F
6	F	F	F	T	T
7	F	F	T	T	T
8	T	F	F	F	T

table 9.

The reader will note that column 5 contains an "F" in rows 3 and 5. These rows may be ignored since when a molecular

sentence of this form is used as a definition, the sentence as a whole is considered true by decision.

Now consider column 3. We see that in rows 1 and 7, p is true. When this is the case, we see from column 4 that '$q \equiv r$' is true also.

This is in accordance with our conception of conditioned definition, namely, that when the condition is fulfilled, that is, p is true, knowledge of the truth-value of q enables us to determine the truth-value of r. Consider again column 3. We see that in rows 2, 4, 6, and 8, p is false. When this is the case, we see from column 4 that '$q \equiv r$' is true in rows 4 and 6 and false in rows 2 and 8. In fact, if we inspect columns 1 and 2, we see that the rows 2, 4, 6, 8 give the four possible combinations of the truth-values of q and r. This means that when the condition is not fulfilled, that is, when p is false, any combination of the truth-values of q and r in '$q \equiv r$' is consistent with the truth of '$p \supset (q \equiv r)$'. This, too, is in accordance with our conception of conditioned definition, namely, that we are unable to say anything about the truth or falsity of r, or the equivalence of q and r, even though we know the truth-value of q, when p is false.

APPENDIX III

Below are the protocol records for the two groups of rats in Experiment III. In the first column is given a rat's identificiation number. Following this number is given the protocol record of this rat's runs into the alley (designated by the letter "A"), returns to the starting path (designated by the letter "S"), and partial choices (designated by the number of the path partially chosen). Thus the following record

A-6-S-4

would mean that the rat ran first into the alley, then to path 6, then to the start, and then to path 4, before it made a final choice. The path finally chosen is indicated in the next column. In the last column is given the total time (in seconds) consumed before the final choice.

PROTOCOL RECORDS OF GROUP F_1

Rat No.		Final Choice	Time In Secs.
207	A-4-5-A-S-2	1	60
32	5-A-S-A-5-4-3-1-A-6-7-8-9-10-A-5-A-S-6-7-8-9-10-A-4-3-2	1	227
146	3-A-1-2-3-A-5-A-S-6-2-1-S-A-5	1	223
201	3-5-A-1-A-S-A-7-8-9-10-6-4-5	3	141
205	2-3-5-4-S-10-8-2-1-S-6-1-S-10-8-6-4-5-3-4-5-9-10-A-1-5-6-7-8-10-S-6-A-10-9-6-A-1-2-S-A-3-4	5	278

Rat No.		Final Choice	Time In Secs.
30	A-S-A-4-3-2-1-S-10-8-6-7-8-9-10-A	5	188
235	A-6-7-S-A-7-6-8-9-10-A-S-5-4-3-2-1-2-1-7	8	167
202	A-S-A-9-10-S-6-7-A-S-4-5-A-1-2-A-1-8-9-10-8-6-10-8-A-5-A-4-5-S-10-A-6-7	8	295
204	A-S-A-6-7-8-9-10-A-3-2-3-1-S-6-A-S-10	9	135
31	9-10-A-7-6-S-A-6-7-8-9-10-S-A-4-5-3-2-1-2-7-6-A-5-A-10	9	225
246	10-9-8-6-7-10-S	10	47
248	A-S-5-4-5-6-7-9-10-A-1-S-A-5-2-3-1-S-10-9-6-A-S-10-7-6-A-S	10	420
52	6-7-5-4-S-1-10-S-A-6-7-S-5-A	10	132
237	A-4-3-2-1-10-9-8-7-6-A-5-S-A-9-8-10-S-1-2-5-6-7-S-A-4-5-4-3-2-1-A-S-10-9-8-7-6-S-1-A-S	10	367
238	A-S-	10	53
53	A-6-7-A-S-A-6-A-1-2-5-4-5-A-7-8-S-A-4-3-A-S-1-2-A-6-8-9	10	444
54	A-3-5-1-A-9-10-9-6-S-4-5-A-7-6-2-3-5-S-A-10-S-4-5-A-10-A-S-1-2-A-S-5-7-8-9	10	161
150	A-6-7-S-A-S-8-A-10-A-3-2-1-2-4-5-S-6-7-8-9-10-A-5-4-3-2-1-S-A-7-8-9-10-S-10-A-6-8-9	10	410
55	A-3-5-1-A-9-10-9-6-S-4-5-A-7-6-2-3-5-S-A-10-S-4-5-A-10-A-S-1-2-A-3-5-7-8-9	10	422
236	1-2-3-4-5-7-6-S-1-10-S-4-3-5-6-7-8-1-S-A-5-3-2-10-A-5-8-A-6-A-3-4-5-A-S	N.C.	
29	10-9-6-A-7-6-A-10-9-S-2-1-S-4-5-3-4-3-6-8-9-10-A-2-1-3-5-6-S-2-5-A-S-6-A-4	N.C.	
200	6-7-S-8-9-8-7-6-5-4-3-S-1-4-5-4-10-S-6-5-S-1-2-1-9-10-7-6-A-5-A-1-A	N.C.	
143	A-6-A-9-10-9-8-A-5-4-A-S-A-3-2-1-6-7-9-10-S-1-A-5-A-S-6-9-10-9-10-S-A-5-1	N.C.	

PROTOCOL RECORDS OF GROUP F_2

Rat No.		Final Choice	Time In Secs.
32	6-4-5-A-S-A-S-1-2-A-5-9-10-7-6-7-6-10-S-3-2-3-4-5-6-A-10-S	1	356
215	A-S-5-4-3	1	57
213	1-2-1-S-5-7-8-9-10-6-7-5-4	1	97
214	4-5-3-2	1	19
220	A-S-A-7-8-7-A-4-5-A-S-A-8-A-S-A-S-1-2-3-A-S-10-7-6-A-2	1	191
25	1-2-3-2	1	40
26	1-3-5-10	1	34
28	A-S-1-2	1	104
33	5-A-1-2-1-S-10-A-S-1-4-5-A-7-6-S	1	158
35	A-S-A-2-3-2	1	125
36	1-2-3-4-5-A-S	1	93
59	5-2-1-S-A-5-4-5-A-S-8-9-10-6-A-4-5	1	135
73	1-5-4-A-S-A-7-6-8-10-A-5-2-1-A-6-S-5-A	1	201
182		1	15
211	1-2-3-4-5-A-6-7-S-A-5-3-2	1	63
52	1	2	14
56	A-5-A-6-7-A-1-A-S-4-5	3	148
31	A-5-6-A-S-1-A-3-9-10-A-9-10-S-A-5-4-3-2 1-2-5	4	277
28	A-5	4	42

Rat No.		Final Choice	Time In Secs.
44	A-7-6-4-5-A-S-x-1-A-8-10-A-S-A-3	4	155
27	A-S-A-4-5	4	63
214	10-9-8-7-6-A-6-7-8-9-10-A-7-8-7-6-A-S	10	174
34	10-9-10-8-7-6-A-S	10	68
208	A-5-4-3-4-A-S-A-8-9-10-6-7-6-3-2-1-A-7-6-8-9-S-1-5-1-S-A-6-A	N.C.	
209	5-1-S-10-8-6-5-4-S-A-5-1-S-10-7-A-5-3-w-1-S-10-9-8-7-6-A-5-4-5-4-3-2-1-S	N.C.	
212	A-5-A-S-A-1-5-6-7-A-S-A-7-8-9-8-6-A-S-10-A-1-3-4-3-5-1-S-A-7-8-7-8-6-8-9-10	N.C.	
30	A-6-7-5-4-S-A-1-2-A-9-10-A-S-6-7-8-6-A-4-5-4-3-2-3-2-1-S-10-9-8-7-6-A-1-A-S-7-8-9-6-A	N.C.	

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